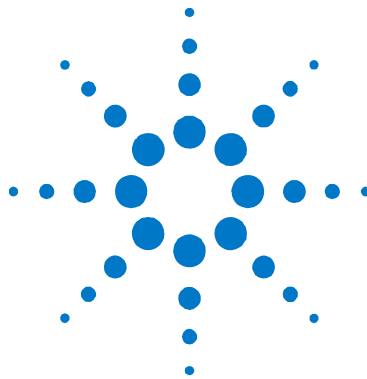




**Agilent Technologies
Modular Power System
and DC Power Analyzer
N6700 and N6705A**

**Programmer's Reference
Guide**



Agilent Technologies

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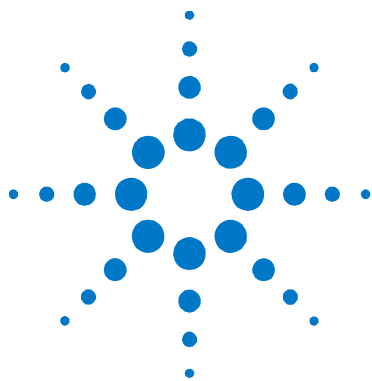
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1

Introduction

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This manual contains reference information to help you program the Agilent N6700 Modular Power System and the Agilent N6705A DC Power Analyzer over the remote interface using the SCPI programming language. The Agilent N6700 and N6705A support the SCPI programming language on all of its remote I/O interfaces.

This manual is a .pdf version of the Programmer's Reference Help file included on the Agilent N6700 and Agilent N6705A Product Reference CD-ROM, which is shipped along with your instrument. This version is provided for customers who require the Programmer's Reference document to be available in .pdf format.

NOTE

You can contact Agilent Technologies at one of the following telephone numbers for warranty, service, or technical support information.

In the United States: (800) 829-4444

In Europe: 31 20 547 2111

In Japan: 0120-421-345

Or use our Web link for information on contacting Agilent in your country or specific location: www.agilent.com/find/assist

Or contact your Agilent Technologies Representative.

The web contains the most up to date version of the manuals. Go to <http://www.agilent.com/find/N6700> to get the latest version of the manuals.

Agilent N6700 and N6705A Product Documentation

The following documents are available on the Agilent N6700 and N6705A Product Reference CD-ROM, as well as from the Agilent Technologies web site at www.agilent.com/find/N6700.

- N6700 and N6705A Programmer's Reference (Help version of this document)
- N6700 User's Guide (for print, order p/n 5969-2939)
N6705A User's Guide (for print order p/n N6705-90000)
- N6700 Service Guide (for print, order p/n 5969-2939)
N6705A Service Guide (for print order p/n N6705-90000)

Instrument Web Server

The Agilent N6700 and N6705A have a built-in Web server that lets you control them directly from an internet browser on your computer. With the Web server, you can access the front panel control functions including the LAN configuration parameters.

NOTE

The built-in Web server only operates over the LAN interface. It requires Internet Explorer 6+ or Netscape 6.2+. You also need the Java (Sun) plug-in. This is included in the Java Runtime Environment. Refer to Sun Microsystem's website <http://www.sun.com>.

The Web server is enabled when shipped. To launch the Web server:

1. Establish a LAN interface connection from your computer to the Agilent N6700 or N6705A.
2. To launch the Web server, enter the instrument's IP address or fully-qualified hostname into the browser's Address field.
3. To begin controlling your instrument, click on the Browser Web Control button in the navigation bar on the left.
4. For additional Help, click on the Help with this Page button.

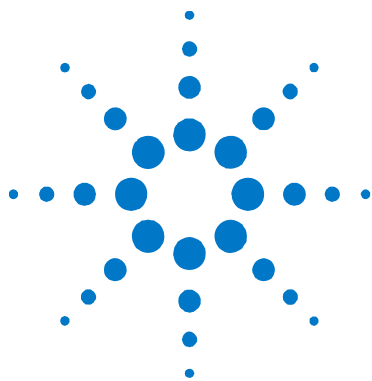
Instrument Drivers and IO Libraries

We have included the following drivers and libraries on the Product Reference CD-ROM shipped with your Agilent N6700 and N6705A.

- IVI-COM Drivers
- VXiplug&play Drivers

Agilent IO Libraries are included on a separate *Automation-Ready CD-ROM*, which is also shipped with your Agilent N6700 and N6705A.

You can also download the drivers and libraries from the Agilent Developer Network at adn.tm.agilent.com.



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Introduction to Programming

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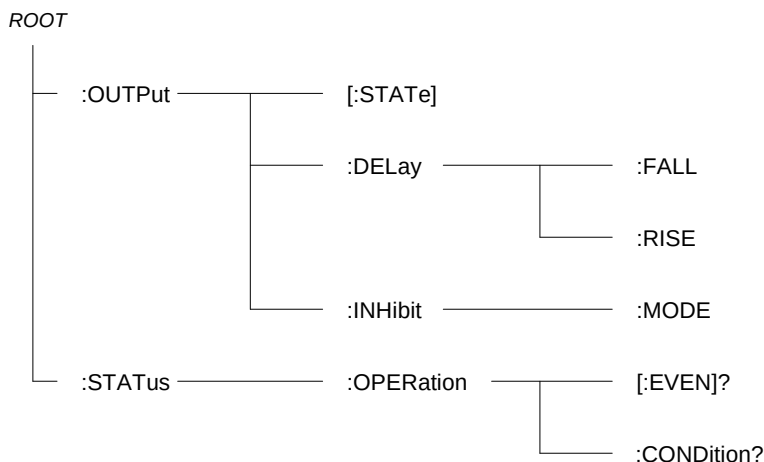
This chapter contains a brief introduction to the SCPI Programming language. SCPI (Standard Commands for Programmable Instruments) is a programming language for controlling test and measurement instruments.

SCPI provides instrument control with a standardized command syntax and style, as well as a standardized data interchange format.

SCPI Commands

SCPI has two types of commands, common and subsystem.

- Common commands generally control overall instrument functions, such as reset, status, and synchronization. All common commands consist of a three-letter mnemonic preceded by an asterisk: *RST *IDN? *SRE 8
- Subsystem commands perform specific instrument functions. They are organized into an inverted tree structure with the "root" at the top. The following figure shows a portion of a subsystem command tree, from which you access the commands located along the various paths.



Multiple Commands in a Message

Multiple SCPI commands can be combined and sent as a single message with one message terminator. There are two important considerations when sending several commands within a single message:

- Use a semicolon to separate commands within a message.
- There is an implied header path that affects how commands are interpreted by the instrument.

The header path can be thought of as a string that gets inserted **before** each command within a message. For the first command in a message, the header path is a null string. For each subsequent command the header path is defined as the characters that make up the headers of the previous command in the message up to and including the last colon separator. An example of a message with two commands is:

```
OUTPut:STATe ON, (@1);PROTection:CLEAr (@1)
```

which shows the use of the semicolon separating the two commands, and also illustrates the header path concept. Note that with the second command, the leading header "OUTPut" was omitted because after the "OUTPut:STATe ON" command, the header path became

defined as "OUTPut" and thus the instrument interpreted the second command as:

```
OUTPut:PROTection:CLEAr (@1)
```

In fact, it would have been syntactically incorrect to include the "OUTPut" explicitly in the second command, since the result after combining it with the header path would be:

```
OUTPut:OUTPut:PROTection:CLEAr (@1)
```

which is incorrect.

Moving Among Subsystems

In order to combine commands from different subsystems, you need to be able to reset the header path to a null string within a message. You do this by beginning the command with a colon (:), which discards any previous header path. For example, you could clear the output protection and check the status of the Operation Condition register in one message by using a root specifier as follows:

```
OUTPut:PROTection:CLEAr (@1);:STATus:OPERation:CONDition? (@1)
```

The following message shows how to combine commands from different subsystems as well as within the same subsystem:

```
VOLTage:LEVel 7.5, (@1);PROTection 10, (@1);:CURRent:LEVel 0.5, (@1)
```

Note the use of the optional header LEVel to maintain the correct path within the subsystems, and the use of the root specifier to move between subsystems.

Including Common Commands

You can combine common commands with subsystem commands in the same message. Treat the common command as a message unit by separating it with a semicolon (the message unit separator). Common commands *do not affect the header path*; you may insert them anywhere in the message.

```
OUTPut OFF, (@1);*RCL 1;OUTPut ON, (@1)
```

Using Queries

Observe the following precautions with queries:

- Add a blank space between the query indicator (?) and any subsequent parameter such as a channel list.
- Allocate the proper number of variables for the returned data.
- Read back all the results of a query before sending another command to the instrument. Otherwise, a *Query Interrupted* error will occur and the unreturned data will be lost.

Coupled Commands

When commands are coupled it means that the value sent by one command is affected by the settings of another command. The following commands are coupled:

- [SOURce:]CURRent and [SOURce:]CURRent:RANGe.
- [SOURce:]VOLTage and [SOURce:]VOLTage:RANGe.

If a range command is sent that places an output on a range with a lower maximum setting than the present level, an error is generated. This also occurs if a level is programmed with a value too large for the present range.

These types of errors can be avoided by sending the both level and range commands as a set, in the same SCPI message. For example,

`CURRent 10, (@1);CURRent:RANGe 10, (@1)<NL>`

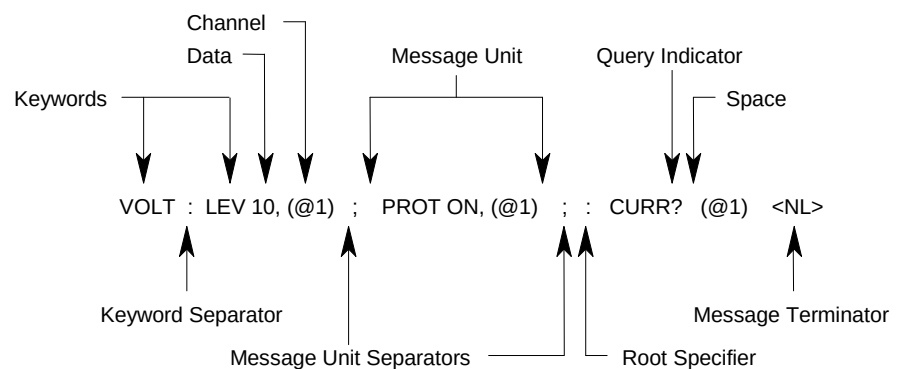
will always be correct because the commands are not executed until the message terminator is received. Because the range and setting information is received as a set, no range/setting conflict occurs.

SCPI Messages

There are two types of SCPI messages, program and response.

- A *program message* consists of one or more properly formatted SCPI commands sent from the controller to the instrument. The message, which may be sent at any time, requests the instrument to perform some action.
- A *response message* consists of data in a specific SCPI format sent from the instrument to the controller. The instrument sends the message only when commanded by a program message "query."

The following figure illustrates the SCPI message structure.



The Message Unit

The simplest SCPI command is a single message unit consisting of a command header (or keyword) followed by a message terminator such as a newline. The message unit may include a parameter after the header. The parameter can be numeric or a string.

```
*RST<NL>
```

```
VOLTage 20, (@1)<NL>
```

Channel List Parameter

The channel parameter is required to address one or more channels. It has the following syntax:

```
(@<channel> [, <channel>] [, <channel>] [, <channel>])
```

You can also specify a range of sequential channels as follows:

```
(@<start_channel>:<end_channel>)
```

For example, (@2) specifies channel 2 and (@1:3) specifies channels 1 through 3. A maximum of 4 channels may be specified through a combination of single channels and ranges. Query results are channel list order-sensitive. Results are returned in the order they are specified in the list.

NOTE

When adding a channel list parameter to a query, you must include a space character between the query indicator (?) and the channel list parameter. Otherwise error –103, Invalid separator will occur.

Headers

Headers, also referred to as keywords, are instructions recognized by the instrument. Headers may be in the long form or in the short form. In the long form, the header is completely spelled out, such as VOLTAGE, STATUS, and DELAY. In the short form, the header has only the first three or four letters, such as VOLT, STAT, and DEL.

When the long form notation is used in this document, the capital letters indicate the equivalent short form. For example, MEASure is the long form, and MEAS indicates the short form equivalent.

Query Indicator

Following a header with a question mark turns it into a query (VOLTage?, VOLTage:TRIGgered?). The ? is the query indicator. If a query contains parameters, place the query indicator at the end of the last header, before the parameters.

```
VOLTage:TRIGgered? MAX, (@1)
```

Message Unit Separator

When two or more message units are combined into a compound message, separate the units with a semicolon.

STATus:OPERation? (@1);QUESTionable? (@1)

Root Specifier

When it precedes the first header of a message unit, the colon becomes the root specifier. It tells the command parser that this is the root or the top node of the command tree.

Message Terminator

A terminator informs SCPI that it has reached the end of a message. The following messages terminators are permitted:

- newline <NL>, which is ASCII decimal 10 or hex 0A.
- end or identify <END> (EOI with ATN false)
- both of the above <NL><END>
- also <CR><NL>

In the examples of this guide, there is an assumed message terminator at the end of each message.

SCPI Conventions and Data Formats

Conventions

The following SCPI conventions are used throughout this guide.

Angle brackets < >	Items within angle brackets are parameter abbreviations. For example, <NR1> indicates a specific form of numerical data.
Vertical bar	Vertical bars separate alternative parameters. For example, VOLT CURR indicates that either "VOLT" or "CURR" can be used as a parameter.
Square brackets []	Items within square brackets are optional. The representation [SOURce:]VOLTage means that SOURce: may be omitted.
Braces { }	Braces indicate parameters that may be repeated zero or more times. It is used especially for showing arrays. The notation <A>{,} shows that parameter "A" must be entered, while parameter "B" may be omitted or may be entered one or more times.
Parentheses ()	Items within parentheses are used in place of the usual parameter types to specify a channel list. The notation (@1:3) specifies a channel list that includes channels 1, 2, and 3. The notation (@1,3) specifies a channel list that includes only channels 1 and 3.

Data Formats

Data programmed or queried from the instrument is ASCII. The data may be numerical or character string.

Numeric and Character Data Formats

Symbol	Description
<NR1>	Digits with an implied decimal point assumed at the right of the least-significant digit. Examples: 273
<NR2>	Digits with an explicit decimal point. Example: 27.3
<NR3>	Digits with an explicit decimal point and an exponent. Example: 2.73E+02
<NRf>	Extended format that includes <NR1>, <NR2> and <NR3>. Examples: 273 27.3 2.73E+02
<NRf+>	Expanded decimal format that includes <NRf> and MIN MAX. Examples: 273 27.3 2.73E+02 MAX. MIN and MAX are the minimum and maximum limit values that are implicit in the range specification for the parameter.
<Bool>	Boolean Data. They can be numeric (0, 1), or named (OFF, ON).
<SPD>	String Program Data. String parameters enclosed in single or double quotes.
<CPD>	Character Program Data. Programs discrete parameters. Accepts both the short form and the long form.
<SRD>	String Response Data. Returns string parameters enclosed in single or double quotes.
<CRD>	Character Response Data. Returns discrete parameters.
<AARD>	Arbitrary ASCII Response Data. Permits the return of undelimited 7-bit ASCII. This data type has an implied message terminator.
<Block>	Arbitrary Block Response Data. Permits the return of definite length and indefinite length arbitrary response data. This data type has an implied message terminator.

Suffixes and Multipliers

Class	Suffix	Unit	Unit with Multiplier
Current	A	ampere	MA (milliampere)
Amplitude	V	volt	MV (millivolt)
Time	S	second	MS (millisecond)
Common Multipliers			
1E3	K	kilo	
1E-3	M	milli	
1E-6	U	micro	

SCPI Command Completion

SCPI commands sent to the instrument are processed either sequentially or in parallel. Sequential commands finish execution before a subsequent command begins. Parallel commands allow other commands to begin executing while the parallel command is still executing.

The following is a list of parallel commands. You should use some form of command synchronization as discussed in this section before assuming that these commands have completed.

OUTPut:STATE	INITiate
VOLTage	OUTPut:PROTEction:CLEAr
CURRent	FUNCTion:MODE

The *WAI, *OPC, and *OPC? common commands provide different ways of indicating when all transmitted commands, including any parallel ones, have completed their operations. Some practical considerations for using these commands are as follows:

***WAI** This command prevents the instrument from processing subsequent commands until all pending operations are completed. For example, the *WAI command can be used to make a voltage measurement after an output on command has completed:

```
OUTPut ON, (@1); *WAI; :MEASure:VOLTage? (@1)
```

***OPC?** This command places a 1 in the Output Queue when all pending operations have completed. Because it requires your program to read the returned value before executing the next program statement, *OPC? can be used to cause the controller to wait for commands to complete before proceeding with its program.

***OPC** This command sets the OPC status bit when all pending operations have completed. Since your program can read this status bit on an interrupt basis, *OPC allows subsequent commands to be executed.

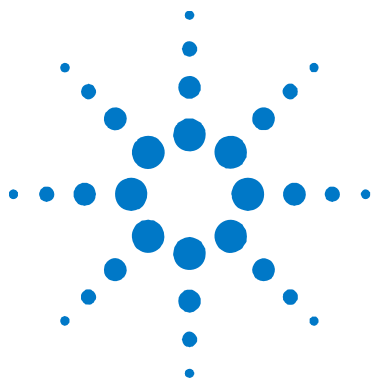
NOTE

The trigger subsystem must be in the Idle state for the status OPC bit to be true. As far as triggers are concerned, OPC is false whenever the trigger subsystem is in the Initiated state.

Device Clear

You can send a Device Clear at any time to abort a SCPI command that may be hanging up the GPIB interface. Device Clear clears the input and output buffers of the instrument and prepares the instrument to accept a new command string. The status registers, error queue, and all configuration states are left unchanged by Device Clear. The following statement shows how to send a device clear over the GPIB interface using *Agilent Basic*:

```
CLEAR 705          IEEE-488 Device Clear
```



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Language Dictionary

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This section gives the syntax and parameters for all the IEEE 488.2 SCPI commands and the Common commands used by the instrument. It is assumed that you are familiar with the material in chapter 2, which explains the terms, symbols, and syntactical structures used here and gives an introduction to programming. You should also be familiar with chapter 4 in the User's Guide, in order to understand how the instrument functions.

Subsystem commands are specific to instrument functions. They can be a single command or a group of commands. The groups are comprised of commands that extend one or more levels below the root. The subsystem commands are arranged alphabetically according to the function they perform.

Common commands are defined by the IEEE 488.2 standard to perform common interface functions. They begin with an * and consist of three letters (command) or three letters and a ? (query). Common commands are grouped along with the subsystem commands according to the function they perform.

SCPI Command Summary

Subsystem Commands

NOTE

Some [optional] commands have been included for clarity. All settings commands have a corresponding query. Not all commands apply to all models.

SCPI Command	Description
ABORt	
:ACQuire (@chanlist)	Resets the measurement trigger system to the Idle state
:DLOG	Stops a running data log (only on N6705A)
:TRANsient (@chanlist)	Resets the transient trigger system to the Idle state
CALibrate	
:CURRent	
[:LEVel] <NRf>, (@channel)	Calibrates the output current programming
:MEASure <NRf>, (@channel)	Calibrates the current measurement
:PEAK (@channel)	Calibrates the peak current limit (only on N675xA/N676xA)
:DATA <NRf>	Enters the calibration value
:DATE <SPD>, (@channel)	Sets the calibration date
:DPRog (@channel)	Calibrates the current downprogrammer
:LEVel P1 P2 P3	Advances to the next calibration step
:PASSword <NRf>	Sets the numeric calibration password
:SAVE	Saves the new cal constants in non-volatile memory
:STATE <Bool> [,<NRf>]	Enables/disables calibration mode
:VOLTage	
[:LEVel] <NRf>, (@channel)	Calibrates the output voltage programming
:CMRR (@channel)	Calibrates common mode rejection ratio (only N675xA/N676xA)
:MEASure <NRf>, (@channel)	Calibrates the voltage measurement
DISPlay	
[:WINDow]:VIEW METER1 METER4	Selects 1-channel or 4-channel meter view
FETCh	(FETCh commands only on N6761A/62A and Opt. 054)
[:SCALar]	
:CURRent [:DC]? (@chanlist)	Returns the average output current
:VOLTage [:DC]? (@chanlist)	Returns the average output voltage
:ARRay	
:CURRent [:DC]? (@chanlist)	Returns the instantaneous output current
:VOLTage [:DC]? (@chanlist)	Returns the instantaneous output voltage
HCOPy	(HCOPy commands only on Agilent N6705A)
:SDUMp:DATA?	Returns an image of the display in .gif format
INITiate	
[:IMMediate]	
:ACQuire (@chanlist)	Enables measurement triggers (only N6761A/62A and Opt. 054)
:DLOG "filename"	Enables the data logger function (only on N6705A)
:TRANsient (@chanlist)	Enables output triggers
:CONTInuous	
:TRANsient <Bool>, (@chanlist)	Enables/disables continuous transient triggers

SCPI Command	Description
MEASure	
[:SCALar]	
:CURRent [:DC]? (@chanlist)	Takes a measurement; returns the average output current
:VOLTage [:DC]? (@chanlist)	Takes a measurement; returns the average output voltage
:ARRay	(ARRay commands only on N6761A/62A and Opt. 054)
:CURRent [:DC]? (@chanlist)	Takes a measurement; returns the instantaneous output current
:VOLTage [:DC]? (@chanlist)	Takes a measurement; returns the instantaneous output voltage
MMEMory	(MMEMory commands only on N6705A)
:ATTRibute? "object", "attribute"	Gets the attributes of a file system object
:DATA[:DEFinite]? "filename"	Gets the file contents; response is a definite length binary block
:DELeTe "filename"	Deletes a file
:EXPort:DLOG "filename"	Exports a data log from the display to a file
OUTPut	
[:STATe] <Bool> [,NORelay], (@chanlist)	Enables/disables the specified output channel(s)
:COUPle	
[:STATe] <Bool>	Enables/disables channel coupling for output synchronization
:CHANNel [<NR1> {,<NR1>}]	Selects which channels are coupled
:DOFFset <NRf>	Specifies a maximum delay offset to synchronize output changes
:MODE AUTO MANual	Specifies the output delay coupling mode (only on N6705A)
:MAX:DOFFset?	Returns the maximum delay offset required for a mainframe
:DELay	
:FALL <NRf+>, (@chanlist)	Sets the output turn-off sequence delay
:RISE <NRf+>, (@chanlist)	Sets the output turn-on sequence delay
:PMODE VOLTage CURRent, (@chanlist)	Sets the mode for turn on/off transitions (only on N6761A/62A)
:INHibit:MODE LATChing LIVE OFF	Sets the remote inhibit input
:PON:STATe RST RCL0	Programs the power-on state
:PROTection	
:CLEAr (@chanlist)	Resets latched protection
:COUPle <Bool>	Enables/disables channel coupling for protection faults
:DELay <NRf+>, (@chanlist)	Sets the over-current protection programming delay
:RELAy:POLarity NORMAl REVerse, (@chanlist)	Sets the output relay polarity (only on Opt. 760)
SENSe	
:CURRent	
[:DC]:RANGe [:UPPer] <NRf+>, (@chanlist)	Selects the current measurement range (only on N6761A/62A)
CCOMPensate <Bool>, (@chanlist)	Enables/disables the capacitive current compensation
:DLOG	(DLOG commands only on N6705A)
:FUNction	
:CURRent <Bool>, (@chanlist)	Enables/disables current data logging
:MINMax <Bool>	Enables/disables min/max data logging
:VOLTage <Bool>, (@chanlist)	Enables/disables voltage data logging
:OFFSet <NR1>	Sets trigger offset as a percent from start of data log duration
:TIME <NRf+>	Sets the duration of the data log in seconds
:TINTerval <NRf+>	Sets the time interval between data log samples
:FUNction "VOLTage" "CURRent", (@chanlist)	Selects the measurement function
:SWEep	(SWEep commands only on N6761A/62A and Opt. 054)
:OFFSet:POINts <NRf+>, (@chanlist)	Defines the trigger offset in the measurement sweep
:POINts <NRf+>, (@chanlist)	Defines the number of data points in the measurement
:TINTerval <NRf+>, (@chanlist)	Sets the measurement sample interval
:VOLTage[:DC]:RANGe [:UPPer] <NRf+>, (@chanlist)	Selects the voltage measurement range (only on N6761A/62A)
:WINDow [:TYPE] HANNing RECTangular, (@chanlist)	Selects the window type (only on N6761A/62A and Opt. 054)

SCPI Command	Description
[SOURce:]	
ARB	(ARB commands only on N6705A)
:COUNT <NRf+> INFIinity, (@chanlist)	Sets the Arb repeat count
:CURRent	
:UDEfined	
:BOSTep[:DATA] <Bool> {,<Bool>}, (@chanlist)	Generate triggers at the Beginning Of STep
:POINts? (@chanlist)	Returns the number of BOST points
:DWELl <NRf> {,<NRf>}, (@chanlist)	Sets the user-defined dwell values
:POINts? (@chanlist)	Returns the number of dwell points
:LEVel <NRf> {,<NRf>}, (@chanlist)	Sets the user-defined current values
:POINts? (@chanlist)	Returns the number of current points
:FUNction STEP RAMP STAIrcase SINusoid PULSe TRAPezoid EXPonential UDVoltage UDCurrent NONE, (@chanlist)	Selects the ARB function
:TERMinate:LAST <Bool>, (@chanlist)	Sets the ARB termination mode
:VOLTage	
:CONVert (@channel)	Converts the selected ARB to a user-defined list
:EXPonential	
:END[:LEVel] < NRf+>, (@channel)	Sets the end voltage of the exponential ARB
:STARt	
[:LEVel] < NRf+>, (@channel)	Sets the initial voltage of the exponential ARB
:TIMe < NRf+>, (@channel)	Sets the length of the start time or delay
:TCONstant < NRf+>, (@channel)	Sets the time constant of the exponential ARB
:TIMe < NRf+>, (@channel)	Sets the time of the exponential ARB
:PULSe	
:END:TIMe < NRf+>, (@channel)	Sets the length of the end time
:STARt	
[:LEVel] < NRf+>, (@channel)	Sets the initial voltage of the pulse
:TIMe < NRf+>, (@channel)	Sets the length of the start time or delay
:TOP	
[:LEVel] < NRf+>, (@channel)	Sets the top level voltage of the pulse
:TIMe < NRf+>, (@channel)	Sets the length of the pulse
:RAMP	
:END	
[:LEVel] < NRf+>, (@channel)	Sets end voltage of the ramp
:TIMe < NRf+>, (@channel)	Sets the length of the end time
:RTIME < NRf+>, (@channel)	Sets the rise time of the ramp
:STARt	
[:LEVel] < NRf+>, (@channel)	Sets the initial voltage of the ramp
:TIMe < NRf+>, (@channel)	Sets the length of the start time or delay
:SINusoid	
:AMPLitude < NRf+>, (@channel)	Sets the amplitude of the sine wave
:FREQuency < NRf+>, (@channel)	Sets the frequency of the sine wave
:OFFSet < NRf+>, (@channel)	Sets the DC offset of the sine wave
:STAIrcase	
:END	
[:LEVel] < NRf+>, (@channel)	Sets the end voltage of the staircase
:TIMe < NRf+>, (@channel)	Sets the length of the end time
:NSTeps < NRf+>, (@channel)	Sets the number of steps in the staircase
:STARt	
[:LEVel] < NRf+>, (@channel)	Sets the initial voltage of the staircase
:TIMe < NRf+>, (@channel)	Sets the length of the start time or delay
:TIMe <NRf+>, (@channel)	Sets the length of the staircase

SCPI Command	Description
[SOURce:]ARB continued	
:STEP	
:END[:LEVel] <NRf+>, (@channel)	Sets the end voltage of the step
:START	
:[:LEVel] <NRf+>, (@channel)	Sets the initial voltage of the step
:TIme <NRf+>, (@channel)	Sets the length of the start time or delay
:TRAPezoid	
:END:TIme <NRf+>, (@channel)	Sets the length of the end time
:FTIme <NRf+>, (@channel)	Sets the length of the fall time
:RTIme <NRf+>, (@channel)	Sets the length of the rise time
:START	
:[:LEVel] <NRf+>, (@channel)	Sets the initial voltage of the trapezoid
:TIme <NRf+>, (@channel)	Sets the length of the start time or delay
:TOP	
:[:LEVel] <NRf+>, (@channel)	Sets the top level voltage of the trapezoid
:TIme <NRf+>, (@channel)	Sets the length of the top of the trapezoid
:UDEfined	
:BOSTep[:DATA] <Bool> {,<Bool>}, (@chanlist)	Generate triggers at the Beginning Of Step
:POINts? (@chanlist)	Returns the number of BOST points
:DWELI <NRf> {,<NRf>}, (@chanlist)	Sets the user-defined dwell values
:POINts? (@chanlist)	Returns the number of dwell points
:LEVel <NRf> {,<NRf>}, (@chanlist)	Sets the user-defined voltage values
:POINts? (@chanlist)	Returns the number of voltage points
CURRent	
:[:LEVel]	
:[:IMMediate][:AMPLitude] <NRf+>, (@chanlist)	Sets the output current
:TRIGgered [:AMPLitude] <NRf+>, (@chanlist)	Sets the triggered output current
:MODE FIXed STEP LIST ARB, (@chanlist)	Sets the current trigger mode
:PROTection	
:DELay[:TIme]<NRf+> (@chanlist)	Sets the over-current protection programming delay
:STARt SCHange CCTRans, (@chanlist)	Sets the over-current protection programming mode
:STATe <Bool>, (@chanlist)	Enables/disables over-current protection on the selected output
:RANGe <NRf+>, (@chanlist)	Sets the output current range (only on N6761A/62A)
DIGital	
:INPut:DATA?	Reads the state of the digital port pins
:OUTPut:DATA <NRf>	Sets the digital port
:PIN<1-7>	
:FUNCTion DIO DINPut TOUTput TINPut FAULt ¹ INHibit ² ONCOuple OFFCouple	Sets the selected pin's function (¹ PIN1 only; ² PIN3 only)
:POLarity POSitive NEGative	Sets the selected pin's polarity
LIST	(LIST commands only on N6761A/62A and Opt. 054)
:COUNT <NRf+> INFinity, (@chanlist)	Sets the list repeat count
:CURRent [:LEVel] <NRf> {,<NRf>}, (@chanlist)	Sets the current list
:POINts? (@chanlist)	Returns the number of current list points
:DWELI <NRf> {,<NRf>}, (@chanlist)	Sets the list of dwell times
:POINts? (@chanlist)	Returns the number of dwell list points
:STEP ONCE AUTO, (@chanlist)	Specifies how the list responds to triggers
:TERMinate:LAST <Bool>, (@chanlist)	Sets the list termination mode
:TOUTput	
:BOSTep[:DATA] <Bool> {,<Bool>}, (@chanlist)	Generate triggers at the Beginning Of STep
:POINts? (@chanlist)	Returns the number of BOST list points
:EOSTep[:DATA] <Bool> {,<Bool>}, (@chanlist)	Generate triggers at the End Of STep
:POINts? (@chanlist)	Returns the number of EOST list points

SCPI Command	Description
[SOURce:]LIST continued	
:VOLTage[:LEVel] <NRf> {,<NRf>}, (@chanlist)	Sets the voltage list
:POINts? (@chanlist)	Returns the number of voltage list points
POWER:LIMit <NRf+>, (@chanlist)	Sets the power limit on output channels
STEP:TOUTput <Bool>, (@chanlist)	Generate a trigger output on the voltage or current step
VOLTage	
[:LEVel]	
[:IMMEDIATE][:AMPLitude] <NRf+>, (@chanlist)	Sets the output voltage
:TRIGgered[:AMPLitude] <NRf+>, (@chanlist)	Sets the triggered output voltage
:MODE FIXed STEP LIST ARB, (@chanlist)	Sets the voltage trigger mode
:PROTection[:LEVel] <NRf+>, (@chanlist)	Sets the over-voltage protection level
:RANGe <NRf+>, (@chanlist)	Sets the output voltage range (only on N6761A/62A)
:SENSe:SOURce INTernal EXTernal, (@chanlist)	Sets the remote sense relays (only on N6705A)
:SLEW[:IMMEDIATE] <NRf+> INfinity, (@chanlist)	Sets the output voltage slew rate
STATus	
:OPERation	
[:EVENT]? (@chanlist)	Returns the value of the operation event register
:CONDition? (@chanlist)	Returns the value of the operation condition register
:ENABle <NRf>, (@chanlist)	Enables specific bits in the Event register
:NTRAnsition <NRf>, (@chanlist)	Sets the Negative transition filter
:PTRAnsition <NRf>, (@chanlist)	Sets the Positive transition filter
:PRESet	Presets all enable and transition registers to power-on
:QUEStionable	
[:EVENT]? (@chanlist)	Returns the value of the questionable event register
:CONDition? (@chanlist)	Returns the value of the questionable condition register
:ENABle <NRf>, (@chanlist)	Enables specific bits in the Event register
:NTRAnsition <NRf>, (@chanlist)	Sets the Negative transition filter
:PTRAnsition <NRf>, (@chanlist)	Sets the Positive transition filter
SYSTem	
:CHANnel	
[:COUNt]?	Returns the number of output channels in a mainframe
:MODEl? (@chanlist)	Returns the model number of the selected channel
:OPTion? (@chanlist)	Returns the option installed in the selected channel
:SERial? (@chanlist)	Returns the serial number of the selected channel
:COMMunicate	
:RLState LOCal REMote RWLock	Specifies the Remote/Local state of the instrument
:TCPip:CONTRol?	Returns the control connection port number
:DATE <yyyy>,<mm>,<dd>	Sets the date of the system clock (only on N6705A)
:ERRor?	Returns the error number and error string
:GROup	
:CATalog?	Returns the groups that have been defined
:DEFine (@chanlist)	Group multiple channels together to create a single output
:DELete <channel>	Removes the specified channel from a group
:ALL	Ungroups all channels
:PASSword:FPANel:RESet	Resets the front panel lock password to zero
:REBoot	Returns the unit to its power-on state
:TIME <hh>,<mm>,<ss>	Sets the time of the system clock (only on N6705A)
:VERsion?	Returns the SCPI version number

SCPI Command	Description
TRIGger	
:ACQuire	(ACQuire commands only on N6761A/62A and Opt. 054)
[:IMMediate] (@chanlist)	Triggers the measurement immediately
:SOURce BUS PIN<1-7> TRANsient<1-4>, (@chanlist)	Sets the measurement trigger source
:ARB	
:SOURce IMMediate EXTernal BUS	Sets the arbitrary waveform trigger source (only on N6705A)
:DLOG	(DLOG commands only on N6705A)
[:IMMediate]	Triggers the data logger immediately
:CURRent	
[:LEVel] <NRf>, (@chanlist)	Sets the current trigger level of the data logger
:SLOPe POSitive NEGative, (@chanlist)	Sets the current trigger slope of the data logger
:SOURce IMMediate EXTernal BUS VOLTage<1-4> CURRent<1-4> ARSKey 000Key	Sets the data logger trigger source
:VOLTage	
[:LEVel] <NRf>, (@chanlist)	Sets the voltage trigger level of the data logger
:SLOPe POSitive NEGative, (@chanlist)	Sets the voltage trigger slope of the data logger
:TRANsient	
[:IMMediate] (@chanlist)	Triggers the output immediately
:SOURce BUS PIN<1-7> TRANsient<1-4>, (@chanlist)	Sets the output trigger source

Common Commands

Command	Description	Command	Description
*CLS	Clear status	*RST	Reset
*ESE <NRf>	Standard event status enable	*SAV <NRf>	Saves an instrument state
*ESR?	Return event status register	*SRE <NRf>	Set service request enable register
*IDN?	Return instrument identification	*STB?	Return status byte
*OPC	Enable "operation complete" bit in ESR	*TRG	Trigger
*OPT?	Return option number	*TST?	Performs self-test, then returns result
*RCL <NRf>	Recalls a saved instrument state	*WAI	Pauses additional command processing until all device commands are done
*RDT?	Return output channel descriptions		

*RST Settings

These settings are set by the *RST (Reset) command

[SOUR:]ARB:COUN	1	[SOUR:]ARB:VOLT:PULS:STAR	MIN
[SOUR:]ARB:CURR:UDEF:BOST	OFF	[SOUR:]ARB:VOLT:PULS:STAR:TIM	0
[SOUR:]ARB:CURR:UDEF:DWEL	0.001	[SOUR:]ARB:VOLT:PULS:TOP	MIN
[SOUR:]ARB:CURR:UDEF:LEV	MIN	[SOUR:]ARB:VOLT:PULS:TOP:TIM	1
[SOUR:]ARB:FUNC	NONE	[SOUR:]ARB:VOLT:RAMP:END	MIN
[SOUR:]ARB:TERM:LAST	OFF	[SOUR:]ARB:VOLT:RAMP:END:TIM	0
[SOUR:]ARB:VOLT:EXP:END	MIN	[SOUR:]ARB:VOLT:RAMP:RTIM	1
[SOUR:]ARB:VOLT:EXP:STAR	MIN	[SOUR:]ARB:VOLT:RAMP:STAR	MIN
[SOUR:]ARB:VOLT:EXP:STAR:TIM	0	[SOUR:]ARB:VOLT:RAMP:STAR:TIM	0
[SOUR:]ARB:VOLT:EXP:TCON	1	[SOUR:]ARB:VOLT:SIN:AMPL	MIN
[SOUR:]ARB:VOLT:EXP:TIM	1	[SOUR:]ARB:VOLT:SIN:FREQ	1
[SOUR:]ARB:VOLT:PULS:END	0	[SOUR:]ARB:VOLT:SIN:OFFS	0

These settings are set by the *RST (Reset) command			
[SOUR:]ARB:VOLT:STA:END	MIN	OUTP	OFF
[SOUR:]ARB:VOLT:STA:END:TIM	0	OUTP:COUP	OFF
[SOUR:]ARB:VOLT:STA:NST	10	OUTP:DEL:FALL	0
[SOUR:]ARB:VOLT:STA:STAR	MIN	OUTP:DEL:RISE	0
[SOUR:]ARB:VOLT:STA:STAR:TIM	0	OUTP:PMOD	VOLT
[SOUR:]ARB:VOLT:STA:TIM	1	OUTP:PROT:COUP	OFF
[SOUR:]ARB:VOLT:STEP:END	MIN	OUTP:PROT:DEL	0.02
[SOUR:]ARB:VOLT:STEP:STAR	MIN	OUTP:REL:POL	NORM
[SOUR:]ARB:VOLT:STEP:STAR:TIM	0	[SOUR:]POW:LIM	MAX
[SOUR:]ARB:VOLT:TRAP:END:TIM	0	SENS:CURR:CCOM	ON
[SOUR:]ARB:VOLT:TRAP:FTIM	1	SENS:CURR:RANG	MAX
[SOUR:]ARB:VOLT:TRAP:RTIM	1	SENS:DLOG:FUNC:CURR	OFF
[SOUR:]ARB:VOLT:TRAP:STAR	MIN	SENS:DLOG:FUNC:MINM	OFF
[SOUR:]ARB:VOLT:TRAP:STAR:TIM	0	SENS:DLOG:FUNC:VOLT	ON
[SOUR:]ARB:VOLT:TRAP:TOP	MIN	SENS:DLOG:OFF	0
[SOUR:]ARB:VOLT:TRAP:TOP:TIM	1	SENS:DLOG:TIME	30
[SOUR:]ARB:VOLT:UDEF:BOST	OFF	SENS:DLOG:TINT	0.1
[SOUR:]ARB:VOLT:UDEF:DWEL	0.001	SENS:FUNC	"VOLT"
[SOUR:]ARB:VOLT:UDEF:LEV	MIN	SENS:SWE:POIN	1024
CAL:STAT	OFF	SENS:SWE:OFFS:POIN	0
[SOUR:]CURR	0.08 or MIN	SENS:SWE:TINT	20.48E-6
[SOUR:]CURR:MODE	FIX	SENS:VOLT:RANG	MAX
[SOUR:]CURR:PROT:DEL	0.02	SENS:WIND	RECT
[SOUR:]CURR:PROT:DEL:MODE	LEG	[SOUR:]STEP:TOUT	FALSE
[SOUR:]CURR:PROT:STAT	OFF	TRIG:ACQ:SOUR	BUS
[SOUR:]CURR:RANG	MAX	TRIG:ARB:SOUR	IMM
[SOUR:]CURR:TRIG	MIN	TRIG:DLOG:CURR	MIN
[SOUR:]DIG:OUTP:DATA	0	TRIG:DLOG:CURR:SLOP	POS
DISP:VIEW	METER1	TRIG:DLOG:SOUR	IMM
INIT:CONT:TRAN	OFF	TRIG:DLOG:VOLT	MIN
[SOUR:]LIST:COUN	1	TRIG:DLOG:VOLT:SLOP	POS
[SOUR:]LIST:CURR	MIN	TRIG:TRAN:SOUR	BUS
[SOUR:]LIST:DWEL	0.001	[SOUR:]VOLT	MIN
[SOUR:]LIST:STEP	AUTO	[SOUR:]VOLT:MODE	FIX
[SOUR:]LIST:TERM:LAST	OFF	[SOUR:]VOLT:PROT	MAX
[SOUR:]LIST:TOUT:BOST	OFF	[SOUR:]VOLT:RANG	MAX
[SOUR:]LIST:TOUT:EOST	OFF	[SOUR:]VOLT:SLEW	9.9E+37
[SOUR:]LIST:VOLT	MIN	[SOUR:]VOLT:TRIG	MIN

Abort Commands

Abort commands cancel any triggered actions.

ABORt:ACQuire (@<chanlist>)

ABORt:TRANsient (@<chanlist>)

These commands cancel any measurement or transient trigger actions presently in process. They return the trigger system back to the Idle state. The two WTG bits in the Status Operation Condition register are also reset.

These commands are executed at power-on and upon execution of *RST.

ABORt : DLOG

This command stops the currently running data log and returns the trigger system back to the Idle state. The data log file is retained.

Only applies to Agilent N6705A.

Calibrate Subsystem

The calibrate subsystem lets you calibrate the instrument. Only one channel can be calibrated at a time. Refer to chapter 3 in the Service Guide for details.

NOTE

If calibration mode has not been enabled with CALibrate:STATE, the calibration commands will generate an error. Use CALibrate:SAVE to save any changes, otherwise all changes will be lost when you exit calibration mode.

CALibrate:CURRENT[:LEVel] <value>, (@<channel>)

This command initiates calibration of the output current. The value that you enter selects the range that is being calibrated.

CALibrate:CURRENT:MEASure <value>, (@<channel>)

This command initiates calibration of the current measurement range. The value that you enter selects the range that is being calibrated.

CALibrate:CURRENT:PEAK (@<channel>)

This command initiates calibration of the peak current limit.

CALibrate:DATA <value>

This command enters a calibration value that you obtain by reading an external meter. You must first select a calibration level (with CALibrate:LEVel) for the value being entered. Data values are expressed in base units - either volts or amperes, depending on which function is being calibrated.

CALibrate:DATE <"date">, (@<channel>) CALibrate:DATE?

This command stores the date that the power module was last calibrated. The calibration date is stored in nonvolatile memory. Enter any ASCII string up to 16 characters. The query returns the date.

NOTE

The firmware does not interpret the string format. The information is not used by the firmware. The command is only provided to store the calibration date.

CALibrate:DPRog (@<channel>)

This command initiates calibration of the current downprogrammer.

CALibrate:LEVel P1|P2|P3

This command is used to advance to the next level in the calibration. **P1** is the first calibration level; **P2** is the second level; **P3** is the third level.

NOTE

Some calibration sequences may require some settling time after sending CAL:LEV but before reading the data from the DVM and sending CAL:DATA.

CALibrate:PASSword <password>

This command lets you change the calibration password. A new password is automatically stored in nonvolatile memory and does not have to be stored with CALibrate:SAVE. If the password is set to 0, password protection is removed and the ability to enter calibration mode is unrestricted. The factory-default password 0 (zero).

CALibrate:SAVE

This command saves calibration constants in non-volatile memory after the calibration procedure has been completed. If calibration mode is exited by programming CALibration:STATe OFF without first saving the new constants, the previous constants are restored.

**CALibrate:STATE ON|OFF [, <password>]
CALibrate:STATE?**

This command enables or disables calibration mode. Calibration mode must be enabled for the instrument to accept any calibration commands. The first parameter specifies the ON (1) or OFF (0) state. The second parameter is the password.

A numeric password is required if calibration mode is being enabled and the existing password is not 0. If the password is not entered or is incorrect, an error is generated and the calibration mode remains disabled. The query returns only the state, not the password.

The *RST value = OFF.

NOTE

When the calibration state is changed from enabled to disabled, new calibration constants are lost unless they have already been stored with CALibrate:SAVE.

CALibrate:VOLTage[:LEVel] <value>, (@<channel>)

This command initiates calibration of the output voltage. The value that you enter selects the range that is being calibrated.

CALibrate:VOLTage:CMRR (@<channel>)

This command initiates calibration of the voltage common mode rejection ratio.

CALibrate:VOLTage:MEASure <value>, (@<channel>)

This command initiates calibration of the voltage measurement range. The value that you enter selects the range that is being calibrated.

Display Commands

The Display commands let you control the front panel display.

DISPlay[:WINDow]:VIEW METER1|METER4 DISPlay[:WINDow]:VIEW?

This command selects the output channel view of the front panel display. METER1 displays one output channel. METER4 displays all output channels up to a maximum of four.

The *RST value = METER1.

HCOPY:SDUMp:DATA?

This query returns the front panel LCD image formatted as a compressed GIF (graphical image file). The size of the compressed image is approximately 5k bytes.

The response is an indefinite length arbitrary binary block of the form:

#0<data-bytes>NL^END

#0 indicates the response is an indefinite length. The end of the block is a newline with EOI asserted

Only applies to Agilent N6705A.

Fetch Subsystem

Fetch commands return a measurement data that has been previously acquired. If you take a voltage measurement, you can fetch only voltage data. If you take a current measurement, you can fetch only current data. Use Fetch commands to return measurement data that has been synchronized with a triggered event.

Agilent Models N6761A and N6762A have simultaneous voltage and current measurement capability. In this case BOTH voltage and current are acquired, regardless of the parameter that is being measured. To return both values of a simultaneous measurement, first use the MEASure command to measure either the output voltage or current. Then use the FETCh command to return the other parameter.

FETCh:ARRay:CURRent[:DC]? (@<chanlist>)
FETCh:ARRay:VOLTage[:DC]? (@<chanlist>)

These queries return an array containing the digitized output current in amperes or output voltage in volts. The data returned by the FETCh command is the result of the last measurement command or acquisition trigger. The data is valid until the next MEASure or INITiate command occurs.

The output current or voltage is digitized whenever a measurement command is sent or an acquisition trigger occurs. The sampling rate is set by SENSE:SWEep:TINTerval. The position of the trigger relative to the beginning of the data buffer is determined by SENSE:SWEep:OFFSet:POINts. The number of points returned is set by SENSE:SWEep:POINts.

These commands are used only with models N6761A/N6762A, or models that have Option 054 (High Speed Test Extensions).

FETCh[:SCALar]:CURRent[:DC]? (@<chanlist>)
FETCh[:SCALar]:VOLTage[:DC]? (@<chanlist>)

These queries return the DC output current in amperes or output voltage in volts. The data returned by the FETCh command is the result of the last measurement command or acquisition trigger. The data is valid until the next MEASure or INITiate command occurs.

These commands are used only with models N6761A/N6762A, or models that have Option 054 (High Speed Test Extensions).

Initiate Subsystem

Initiate commands initialize the trigger system. This enables the trigger system to receive triggers.

INITiate[:IMMediate]:ACQuire (@<chanlist>)
INITiate[:IMMediate]:TRANSient (@<chanlist>)

These commands control the enabling of both measurement and transient triggers. When a trigger is initiated, an event on a selected trigger source causes the specified triggering action to occur. If the trigger system is not initiated, all triggers are ignored.

It takes a few milliseconds for the power supply to be ready to receive a trigger signal after receiving the INITiate command, and it can take longer if the value of SENSE:SWEep:OFFSet:POINts is negative. If a trigger occurs before the trigger system is ready for it, the trigger will be ignored. The user can test the WTG_MEAS bit in the operation status register to know when the power supply is ready to receive a trigger after initiating.

INITiate[:IMMediate]:DLOG <"filename">

This command starts a data log. The filename in which to save the data should be the full path and filename. All data log files must have the extension .dlog.

Only applies to Agilent N6705A.

INITiate:CONTinuous:TRANSient ON|OFF, (@<chanlist>)
INITiate:CONTinuous:TRANSient? (@<chanlist>)

This command continuously initiates the output trigger system. The enabled state is ON (1); the disabled state is OFF (0). With continuous triggering disabled, the output trigger system must be initiated for each trigger using the INITiate:TRANSient command.

The *RST value = OFF.

Measurement Subsystem

Measure commands measure the output voltage or current. They trigger the acquisition of new data before returning the reading. Measurements are performed by digitizing the instantaneous output voltage or current for a specified measurement time, storing the results in a buffer, and calculating the average value. Use Measure commands when the measurement does not need to be synchronized with any other event.

Agilent Models N6761A and N6762A have simultaneous voltage and current measurement capability. In this case BOTH voltage and current are acquired, regardless of the parameter that is being measured. To return both values of a simultaneous measurement, first use the MEASure command to measure either the output voltage or current. Then use the FETCh command to return the other parameter.

MEASure:ARRAY:CURRENT[:DC]? (@<chanlist>)

MEASure:ARRAY:VOLTage[:DC]? (@<chanlist>)

These queries initiate and trigger a measurement and return an array containing the digitized output current in amperes or output voltage in volts.

The output voltage or current is digitized whenever a measurement command is sent or an acquisition trigger occurs. The sampling rate is set by SENSE:SWEep:TINTerval. The position of the trigger relative to the beginning of the data buffer is determined by SENSE:SWEep:OFFSet:POINTs. The number of points returned is set by SENSE:SWEep:POINTs.

These commands are used only with models N6761A/N6762A, or models that have Option 054 (High Speed Test Extensions).

MEASure[:SCALar]:CURRENT[:DC]? (@<chanlist>)

MEASure[:SCALar]:VOLTage[:DC]? (@<chanlist>)

These queries initiate and trigger a measurement and return the DC output current in amperes or output voltage in volts.

The default time interval and number of points settings yield a measurement time of 21 milliseconds per reading (1024 data points at 20.48 microsecond intervals). The default windowing function is Rectangular.

The time interval, number of points, and windowing function can be programmed on models N6761A/N6762A, and on models that have Option 054 (High Speed Test Extensions).

Memory Subsystem

Memory commands provide mass storage capability for output measurement data. The mass storage may be either internal or external to the instrument. The mass storage device is selected using the full path embedded within the filename. The filename parameter is a string.

MMEMory:ATTRibute? <"object">, <"attribute">

This query returns the attributes of a file system drive or of an individual file.

"INTERNAL" This object specifies the instrument's internal drive.

"EXTERNAL" This object specifies the instrument's USB drive.

"<filename>" The filename object should be the full path and filename.

"DriveCapacity" This attribute specifies the capacity of the internal or external drive in bytes. Only applies when the object is INTERNAL: or EXTERNAL:

"DriveFreeSpace" This attribute specifies the number of free bytes on the internal or external drive. Only applies when the object is INTERNAL: or EXTERNAL.

"FileSize" This attribute specifies the size of the file or folder. Only applies when the object is <filename>.

Only applies to Agilent N6705A.

MMEMory:DATA[:DEFinite]? <"filename">

This query returns the contents of the specified file. The filename should be the full path and filename. The response is a definite length arbitrary binary block of the form: #<D><DD...D><data-bytes>

#<D> indicates the response is definite length, with <D> the number of digits are in the length field. <DD...D> indicates the length of the binary block containing the actual file. <data-bytes> are the contents of the file.

Only applies to Agilent N6705A.

MMEMory:DELeTe <"filename">

This command deletes the specified file. The filename should be the full path and filename. All data log files must have the extension .dlog

Only applies to Agilent N6705A.

MMEMory:EXPort:DLOG <"filename">

This command exports a data log file. The export filename should be the full path and filename.

Only applies to Agilent N6705A.

Output Subsystem

The Output subsystem controls the output, power-on, protection, and relay functions.

OUTPut[:STATe] ON|OFF [,NORelay], (@<chanlist>)
OUTPut[:STATe]? (@<chanlist>)

This command enables or disables the specified output channel(s). The enabled state is ON (1); the disabled state is OFF (0). The state of a disabled output is a condition of zero output voltage and zero source current. If output and sense relays are installed (Option 761), the relays will open when the output is disabled and close when the output is enabled. The query returns 0 if the output is off, and 1 if the output is on.

The optional NORelay parameter lets you turn the output state on or off and leave the state of the relays unchanged. When not specified, the relays open and close as the output is turned off and on. Relays only open and close when the output is at zero voltage; zero current.

Separate delays can be programmed for the off-to-on and the on-to-off transition using OUTPut:DElay:RISE and OUTPut:DElay:FALL.

The *RST value = OFF.

NOTE

Because of internal circuit start-up procedures and any installed relay options, the output on command may take between 35 and 50 milliseconds to complete its function. Conversely, the output off command may take between 20 and 25 milliseconds to complete its function. To mitigate this built-in delay, you can program the output to zero volts rather than using the output on/off command.

OUTPut[:STATe]:COUPle[:STATe] ON|OFF
OUTPut[:STATe]:COUPle[:STATe]?

This command enables or disables output coupling or synchronization between multiple output channels. When output channels have been coupled and the Digital connector pins have been properly configured, enabling this function allows you to control the output state of multiple outputs in multiple mainframes by controlling the output state of one output channel. The query returns 1 if outputs are coupled, and 0 if outputs are not coupled.

When this command is sent, all output channels in the mainframe will go to the output Off state.

This setting is non-volatile and not affected by *RST.

Does not apply to Agilent N6705A as the function is always enabled.

**OUTPut[:STATE]:COUPle:CHANnel [<channel> {,<channel>}]
OUTPut[:STATE]:COUPle:CHANnel?**

This command specifies which output channels in the mainframe are controlled by the output synchronization function. The output channels that have been synchronized or coupled will turn on and off together when any one of them is turned on or off, or when a signal is received from a digital connector pin that has been configured as an On couple or an Off couple pin. The query returns the channels that are coupled. Multiple responses are separated by commas.

There can be only one set of coupled channels in a mainframe. If another coupled set is programmed, the existing set is deleted and the new set applies. If you do not specify an output channel, this command will uncouple all output channels on the mainframe

When this command is sent, all output channels in the mainframe will go to the output Off state.

This setting is non-volatile and not affected by *RST.

**OUTPut[:STATE]:COUPle:DOFFset <delay>
OUTPut[:STATE]:COUPle:DOFFset?**

This command specifies a maximum delay offset to synchronize the output state changes. Units are in seconds. Setting this time to the maximum delay offset specified for any output channel that is being coupled will cause all of the output channels to synchronize to the turn-on times specified by OUTput:DElay:RISE. This delay only applies to coupled channels, provided OUTput:COUPle is enabled.

When this command is sent, all output channels in the mainframe will go to the output Off state.

This setting is non-volatile and not affected by *RST.

**OUTPut[:STATE]:COUPle:DOFFset:MODE AUTO|MANual
OUTPut[:STATE]:COUPle:DOFFset:MODE?**

This command sets the mode that determines where the output coupling delay value comes from when outputs are synchronized.

The output synchronization function lets you select specific outputs to be synchronized as well as **manually** specify the delay offset, which serves as a reference for the user-programmed turn-on delays. Refer to Appendix D in the User's Guide for detailed information.

AUTO The N6705A automatically calculates the maximum delay offset value for all the output channels in the mainframe.

MANual The N6705A uses the delay offset value set by OUTPut:COUPle:DOFFSet.

This setting is non-volatile and not affected by *RST.

Only applies to Agilent N6705A.

OUTPut[:STATE]:COUPLe:MAX:DOFFset?

This query returns the largest delay offset that is required for the output channels installed in this mainframe. Refer to Appendix D in the User's Guide for detailed information.

OUTPut[:STATE]:DElay:FALL <delay>|MIN|MAX, (@<chanlist>)
OUTPut[:STATE]:DElay:FALL? [MIN|MAX,] (@<chanlist>)

This command sets the delay in seconds that the instrument waits before disabling the specified output. It affects on-to-off transitions including changes in the OUTPut:STATE as well as transitions due to changes in the voltage range or current range. It does NOT affect transitions to off caused by protection functions. Delay times can range from 0 to 1.023 seconds in increments of 1 millisecond.

This command allows multiple output channels to turn off in sequence. Each output will not turn off until its delay time has elapsed.

The *RST value = 0.

OUTPut[:STATE]:DElay:RISE <delay>|MIN|MAX, (@<chanlist>)
OUTPut[:STATE]:DElay:RISE? [MIN|MAX,] (@<chanlist>)

This command sets the delay in seconds that the instrument waits before enabling the specified output. It affects all off-to-on transitions including changes in the OUTPut:STATE as well as transitions due to OUTPut:PROTection:CLEar. Delay times can range from 0 to 1.023 seconds in increments of 1 millisecond.

This command allows multiple output channels to turn on in sequence. Each output will not turn on until its delay time has elapsed.

The *RST value = 0.

NOTE

Output channel turn-on characteristics vary across the three module types - DC Power, Autoranging, and Precision (refer to Appendix D in the User's Guide for more information about turn-on synchronization). When several channels of the **same** module type are programmed off-to-on by this command, output sequencing is precisely determined by the programmed turn-on delays.

However, when outputs of **different** module types are sequenced using this command, there may be an additional offset of a few milliseconds from one output to another. This offset is the same for each module type and is repeatable. Once you have characterized this offset, using an oscilloscope for example, you can adjust the programmed delays to compensate for the offset and give the desired output sequencing.

Outputs within the same module type can also have an offset if one model has output relays (Option 761) and another does not. These offsets are also repeatable and can be compensated by adjusting the programmed delay values.

OUTPut[:STATe]:PMODE VOLTage|CURRent, (@<chanlist>)
OUTPut[:STATe]:PMODE? (@<chanlist>)

This command sets the preferred mode for output on or output off transitions on Agilent N6761A/N6762A power modules only. This command allows output state transitions to be optimized for either constant voltage or constant current operation. Turn-on and turn-off overshoots are minimized for the preferred mode of operation.

VOLTage Minimizes output on/off voltage overshoots in constant voltage operation.

CURRent Minimizes output on/off current overshoots in constant current operation.

The *RST value = VOLTage.

OUTPut:INHibit:MODE LATChing|LIVE|OFF
OUTPut:INHibit:MODE?

This command selects the mode of operation of the Inhibit input (INH). The inhibit function shuts down ALL output channels in response to an external signal on the Inhibit input. If an output channel has been turned off by OUTPut:STATe, the inhibit function does not affect the output channel while it is in the OFF state. The Inhibit mode setting is stored in non-volatile memory. The following modes can be selected:

LATChing Causes a logic-true transition on the Inhibit input to disable all outputs. The outputs remain disabled until the Inhibit input is returned to logic-false and the latched INH status bit is cleared by sending the OUTP:PROT:CLE command or a protection clear command from the front panel.

LIVE Allows the enabled outputs to follow the state of the Inhibit input. When the Inhibit input is true, the outputs are disabled. When the Inhibit input is false, the outputs are re-enabled.

OFF The Inhibit input is ignored.

OUTPut:PON:STATe RST|RCL0
OUTPut:PON:STATe?

This command determines if the power-on state is set to the *RST (RST) state or the instrument state stored in memory location 0 (RCL0). The parameter is saved in non-volatile memory. Instrument states can be stored using the *SAV command.

As shipped, the *RST command is issued at power on.

OUTPut:PROTection:CLEAr (@<chanlist>)

This command clears the latched protection status that disables the output when an over-voltage, over-temperature, over-current, power-fail, or Inhibit status condition is detected. All conditions that generate the fault must be removed before the latched status can be cleared. The output is restored to the state it was in before the fault condition occurred.

NOTE

If a protection shutdown occurs during an output list, the list continues running even though the output is disabled. When the protection status is cleared and the output becomes enabled again, the output will be set to the values of the step that the list is presently at.

OUTPut:PROTection:COUPle ON|OFF
OUTPut:PROTection:COUPle?

This command enables/disables output coupling for protection faults. When enabled, ALL output channels are disabled when a protection fault occurs on any output channel. The enabled state is On (1); the disabled state is Off (0). When disabled, only the affected output channel is disabled when a protection fault is triggered.

The *RST value = OFF.

OUTPut:PROTection:DELAy <delay>|MIN|MAX, (@<chanlist>)
OUTPut:PROTection:DELAy? [MIN|MAX,] (@<chanlist>)

This command sets the over-current protection programming delay. This prevents momentary changes in status that can occur during *programming* from triggering the over-current protection function. Programmed values can range from 0 to 255 milliseconds.

The *RST value = 20 ms.

OUTPut:RELAy:POLarity NORMal|REVerse, (@<chanlist>)
OUTPut:RELAy:POLarity? (@<chanlist>)

This command sets the output polarity on power modules with polarity reversal relays installed (Option 760). Polarity reversal affects both the output and the sense terminals.

NORMal Causes the output polarity to be the **same** as the output connector labeling.

REVerse Causes the output polarity to be the **reverse** of the output connector labeling. Note that this command briefly turns the output off while the output and sense terminal polarities are switched.

The *RST value = NORMal.

NOTE

When installed on models N6742B and N6773A, Option 760 (Polarity Reversal) will limit the output current of these models to a maximum of 10 A. Option 760 is not available on models N6741B, N6751A/N6752A, or N6761A/N6762A

Sense Subsystem

Sense commands control the current measurement range, the bandwidth detector of the instrument, and the data acquisition sequence.

SENSe:CURRent[:DC]:RANge[:UPPer] <value>|MIN|MAX, (@<chanlist>)
SENSe:CURRent[:DC]:RANge[:UPPer]? [MIN|MAX,] (@<chanlist>)

This command selects a DC current measurement range on models that have multiple ranges. The value that you enter must be higher than the maximum current that you expect to measure. Units are in amperes. The instrument selects the range with the best resolution for the value entered. MIN selects the lowest range; MAX selects the highest range.

When queried, the returned value is the maximum DC current that can be measured on the range that is presently set. Refer to appendix A in the User's Guide for the available ranges for each model.

The *RST value = the highest available range.

SENSe:CURRent:CCOMpensate ON|OFF, (@<chanlist>)
SENSe:CURRent:CCOMpensate? (@<chanlist>)

This command turns current measurement compensation on or off on the specified output channels.

When turned ON, this function corrects the output current measurement for the current that flows into the output capacitor. This current is only significant when high dv/dt occurs, and because this current does not flow into the output load, it represents an error when not corrected. Note that turning on the current measurement compensation increases the peak to peak noise in current .

The *RST value = ON.

SENSe:DLOG:FUNction:CURRent ON|OFF, (@<chanlist>)
SENSe:DLOG:FUNction:CURRent? (@<chanlist>)

This command turns current data logging on or off on the specified output channels.

The *RST value = OFF.

Only applies to Agilent N6705A.

SENSe:DLOG:FUNction:VOLTage ON|OFF, (@<chanlist>)
SENSe:DLOG:FUNction:VOLTage? (@<chanlist>)

This command turns voltage data logging on or off on the specified output channels.

The *RST value = ON.

Only applies to Agilent N6705A.

SENSe:DLOG:FUNCTION:MINMax ON|OFF
SENSe:DLOG:FUNCTION:MINMax?

This command enables/disables the logging of the minimum and maximum values to the data log file.

Although data is logged to the data file at the logging interval, it is sampled at a rate of 20.48 microseconds. When the data is logged, the minimum, maximum, and average values are computed from all of the samples taken during the logging interval. When disabled, only the average values are entered into the data log file. When enabled, all three values are entered into the data log file - thus tripling its size.

This function cannot be enabled when interleaved data logging is in effect.

The *RST value = OFF.

Only applies to Agilent N6705A.

SENSe:DLOG:OFFset <percent>
SENSe:DLOG:OFFset?

This command specifies the data log trigger offset as a percent of the total data log duration. Programming a trigger offset lets you specify the percent of pre-trigger data that will be logged to the data log file.

A percent of 0 means the trigger occurs at the beginning of the running data log, while 100 means that the trigger occurs at the end of the running data log. Any value between 0 and 100 can be set.

This function cannot be set to values other than zero when interleaved data logging is in effect.

The *RST value = 0 (zero).

Only applies to Agilent N6705A.

SENSe:DLOG:TIME <seconds>
SENSe:DLOG:TIME?

This command specifies the duration of the data log in seconds. Values from 1 second up to the equivalent of 99,999 hours can be specified.

The resulting data log file cannot exceed 2E9 bytes (1.87 Gbytes in Microsoft Windows units). If the duration value causes the file to exceed this size, the time interval will automatically be adjusted to a longer interval to reduce the file size.

The *RST value = 30.

Only applies to Agilent N6705A.

**SENSe:DLOG:TINterval <seconds>
SENSe:DLOG:TINterval?**

This command specifies the time interval between data log samples in seconds. Values from 1 millisecond to 60 seconds can be specified.

This function cannot be set to values less than 75 milliseconds when interleaved data logging is in effect.

The *RST value = 0.1.

Only applies to Agilent N6705A.

**SENSe:FUNCTION "VOLTage"|"CURRent", (@<chanlist>)
SENSe:FUNCTION? (@<chanlist>)**

This command selects a measurement function on models that do not have simultaneous voltage and current measurement capability. This command is required so that the acquisition system knows which measurement function to acquire when a measurement is triggered.

The *RST value = "VOLTage".

**SENSe:SWEep:OFFSet:POINts <points>|MIN|MAX, (@<chanlist>)
SENSe:SWEep:OFFSet:POINts? [MIN|MAX,] (@<chanlist>)**

This command defines the offset in a data sweep when an acquire trigger is used on models that have measurement controls. Programmed values can range from -4095 through 2,000,000,000 (2E9). Negative values represent data samples taken prior to the trigger. Positive values represent the delay after the trigger occurs but before the samples are acquired.

The *RST value = 0.

**SENSe:SWEep:POINts <points>|MIN|MAX, (@<chanlist>)
SENSe:SWEep:POINts? [MIN|MAX,] (@<chanlist>)**

This command defines the number of points in a measurement on models that have measurement controls. Programmed values can range from 1 to 4096.

The *RST value = 1024.

**SENSe:SWEep:TINterval <interval>|MIN|MAX, (@<chanlist>)
SENSe:SWEep:TINterval? [MIN|MAX,] (@<chanlist>)**

This command defines the time period between samples in seconds on models that have measurement controls. Programmed values can range from 20.48 microseconds to 40000 seconds. Values are rounded to the nearest 20.48 microsecond increment.

Note that the query returns the time period rounded to the nearest 20.48 microsecond increment; which is not necessarily the value that was programmed.

The *RST value = 20.48 microseconds.

SENSe:VOLTage[:DC]:RANGE[:UPPer] <value>|MIN|MAX, (@<chanlist>)
SENSe:VOLTage[:DC]:RANGE[:UPPer]? [MIN|MAX,] (@<chanlist>)

This command selects a DC voltage measurement range on models that have multiple ranges. The programmed value must be the maximum voltage that you expect to measure. Units are in volts. The instrument selects the range with the best resolution for the value entered. MIN selects the lowest range; MAX selects the highest range.

When queried, the returned value is the maximum DC voltage that can be measured on the range that is presently set. Refer to Appendix A in the User's Guide for the available ranges for each model.

The *RST value = the highest available range.

SENSe:WINDow[:TYPE] HANNing|RECTangular, (@<chanlist>)
SENSe:WINDow[:TYPE]? (@<chanlist>)

This command sets the window function used in DC measurement calculations on models that have measurement controls. Select from:

HANNing A signal conditioning window that reduces errors in DC measurement calculations in the presence of periodic signals such as AC line ripple. The Hanning window multiplies each point in the sample by the function $\cos^4$.

RECTangular A window that returns measurement calculations with no signal conditioning.

Note that neither window function alters the instantaneous voltage or current data returned in the measurement array.

The *RST value = RECTangular.

Source Subsystem

The Source subsystem programs the ARB, current, digital, list, step, and voltage functions. RANGE and LIST commands do not apply to all models. Refer to chapter 1 in the User's Guide under "Model Differences". ARB commands apply only to Model N6705A.

[SOURCE:]ARB:COUNT <count>|MIN|MAX|INF, (@<chanlist>)
[SOURCE:]ARB:COUNT? [MIN|MAX,] (@<chanlist>)

This command sets the number of times that the arbitrary waveform is repeated. The range is 1 through 256. Use INFINITY to execute a list indefinitely. Use ABORT:TRANSient to stop the ARB.

The *RST value = 1

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:CURRENT:UDEFINED:BOSTep[:DATA] ON|OFF {,ON|OFF}, (@<chanlist>)
[SOURCE:]ARB:CURRENT:UDEFINED:BOSTep[:DATA]? (@<chanlist>)

This command specifies which user-defined ARB points or steps will generate a trigger out signal at the beginning of the step (BOSTep). A comma-delimited list of up to 512 steps may be programmed. The state is either ON (1) or OFF (0). A trigger is only generated when the state is set to ON.

The *RST value = 1 step with a value of OFF.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:CURRENT:UDEFINED:BOSTep:POINTS? (@<chanlist>)

This query returns the number of beginning-of-step points in the user-defined ARB, not the point values.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:CURRENT:UDEFINED:DWELL <time> {,<time>}, (@<chanlist>)
[SOURCE:]ARB:CURRENT:UDEFINED:DWELL? (@<chanlist>)

This command specifies the dwell time for each user-defined ARB point. A comma-delimited list of up to 512 points may be programmed. Dwell time is the time that the output will remain at a specific point. Dwell times can be programmed from 0 to 262.144 seconds with the following resolution:

Range	Resolution	Range	Resolution
0 to 0.262144	1 microsecond	0 to 26.2144	100 microseconds
0 to 2.62144	10 microseconds	0 to 262.144	1 millisecond

At the end of the dwell time, the output state of the unit depends upon the ARB:TERM:LAST program settings. See ARB:TERM:LAST

The *RST value = 1 point with a value of 0.001.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:CURRENT:UDEFined:DWELL:POINTS? (@<chanlist>)

This query returns the number of dwell points in the user-defined ARB, not the point values.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:CURRENT:UDEFined:LEVel <current> {,<current>}, (@<chanlist>)
[SOURce:]ARB:CURRENT:UDEFined:LEVel? (@<chanlist>)

This command specifies the current setting for each user-defined ARB point in amperes. A comma-delimited list of up to 512 points may be programmed.

The *RST value = 1 point set MIN.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:CURRENT:UDEFined:LEVel:POINTS? (@<chanlist>)

This query returns the number of current points in the user-defined ARB, not the point values.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:FUNCTION <function>, (@<chanlist>)
[SOURce:]ARB:FUNCTION? (@<chanlist>)

This command sets the function of the arbitrary waveform generator.

STEP	Specifies a voltage step
RAMP	Specifies a voltage ramp
STAircase	Specifies a voltage staircase
SINusoid	Specifies a voltage sine wave
PULse	Specifies a voltage pulse
TRAPezoid	Specifies a voltage trapezoid
EXPonential	Specifies an exponential voltage
UDVoltage	Specifies a user-defined voltage waveform
UDCurrent	Specifies a user-defined current waveform
NONE	Specifies no arbitrary waveform

Use [SOUR:]VOLT:MODE ARB and [SOUR:]CURR:MODE ARB to enable the instrument to generate arbitrary voltage or current waveforms when the transient system is initiated and triggered.

The *RST value = NONE.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:TERMinate:LAST ON|OFF, (@<chanlist>)
[SOURCE:]ARB:TERMinate:LAST? (@<chanlist>)

This command determines the output value when the ARB terminates. The state is either ON (1) or OFF (0). When ON, the output voltage or current remains at the last ARB value. The last voltage or current value becomes the IMMEDIATE value when the ARB completes. When OFF, and also when the ARB is aborted, the output returns to the settings that were in effect before the ARB started.

The *RST value = OFF.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:CONVert (@<chanlist>)

This command converts the specified arbitrary waveform to a list of points in the user-defined voltage waveform format. This makes it possible to modify/edit one of the "standard" arbitrary waveforms. The command performs the same function as the "Edit Points" button on the front panel ARB Properties dialog windows.

To use this command you must first specify one of the following ARB functions to convert using [SOURCE:]ARB:FUNCTION: STEP, RAMP, STAIRcase, SINusoid, PULse, TRAPezoid, or EXPonential

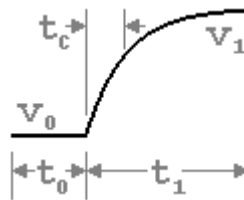
If the ARB function of the specified channel is already set to a user-defined voltage waveform or set to NONE, the command is ignored and the conversion is not performed.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:EXPonential:END[:LEVel] <voltage>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:EXPonential:END[:LEVel]? [MIN|MAX,](@<chanlist>)

This command specifies the end level voltage (after the exponential waveform ends) in volts. Referenced to V_1 in the Exponential diagram.

Exponential diagram



The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:EXPonential:START[:LEVel] <volt>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:EXPonential:START[:LEVel]? [MIN|MAX,](@<chanlist>)

This command specifies the initial voltage (before the exponential waveform starts) in volts. Referenced to V_0 in the Exponential diagram.

The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:EXPonential:START:TIME <time>|MIN|MAX, (@<chanlist>)
 [SOURce:]ARB:VOLTage:EXPonential:START:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the delay in seconds after the trigger is received, but before the exponential waveform occurs. Start times can be programmed from 0 to 262.143 seconds. Referenced to t_0 in the Exponential diagram.

The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:EXPonential:TCONstant <time>|MIN|MAX, (@<chanlist>)
 [SOURce:]ARB:VOLTage:EXPonential:TCONstant? [MIN|MAX,](@<chanlist>)

This command specifies the time constant of the curve. Time constants can be programmed from 0 to 262.143 seconds. Referenced to t_c in the Exponential diagram.

The *RST value = 1

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:EXPonential:TIME <time>|MIN|MAX, (@<chanlist>)
 [SOURce:]ARB:VOLTage:EXPonential:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the total time for the voltage to go from the starting level to the ending level in seconds. Times can be programmed from 0 to 262.143 seconds. Referenced to t_1 in the Exponential diagram.

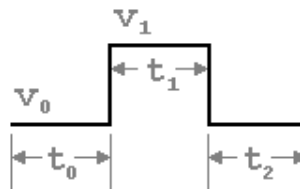
The *RST value = 1

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:PULSe:END:TIME <time>|MIN|MAX, (@<chanlist>)
 [SOURce:]ARB:VOLTage:PULSe:END:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the time in seconds, after the pulse completes, that the start-level voltage persists. End times can be programmed from 0 to 262.143 seconds. Referenced to t_2 in the Pulse diagram.

Pulse diagram



The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:PULSe:START[:LeVel] <voltage>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:PULSe:START[:LeVel]? [MIN|MAX,](@<chanlist>)

This command specifies the voltage before and after the pulse occurs in volts. Referenced to V_0 in the Pulse diagram.

The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:PULSe:START:TIME <time>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:PULSe:START:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the delay in seconds after the trigger is received, but before the pulse occurs. Start times can be programmed from 0 to 262.143 seconds. Referenced to t_0 in the Pulse diagram.

The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:PULSe:TOP[:LeVel] <voltage>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:PULSe:TOP[:LeVel]? [MIN|MAX,](@<chanlist>)

This command specifies the top level of the pulse in volts. Referenced to V_1 in the Pulse diagram.

The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:PULSe:TOP:TIME <time>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:PULSe:TOP:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the time of the pulse in seconds. Top times can be programmed from 0 to 262.143 seconds. Referenced to t_1 in the Pulse diagram.

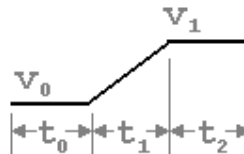
The *RST value = 1

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:RAMP:END[:LeVel] <voltage>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:RAMP:END[:LeVel]? [MIN|MAX,](@<chanlist>)

This command specifies the end level voltage (after the ramp occurs) in volts. Referenced to V_1 in the Ramp diagram.

**Ramp
diagram**



The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:RAMP:END:TIME <time>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:RAMP:END:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the time in seconds, after the ramp completes, that the end-level voltage persists. End times can be programmed from 0 to 262.143 seconds. Referenced to t_2 in the Ramp diagram.

The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:RAMP:RTIME <time>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:RAMP:RTIME? [MIN|MAX,](@<chanlist>)

This command specifies the rise time of the ramp in seconds. Rise times can be programmed from 0 to 262.143 seconds. Referenced to t_1 in the Ramp diagram.

The *RST value = 1

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:RAMP:START[:LEVEL] <voltage>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:RAMP:START[:LEVEL]? [MIN|MAX,](@<chanlist>)

This command specifies the initial voltage (before the ramp occurs) in volts. Referenced to V_0 in the Ramp diagram.

The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:RAMP:START:TIME <time>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:RAMP:START:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the time in seconds, after the trigger is received, but before the ramp occurs. Start times can be programmed from 0 to 262.143 seconds. Referenced to t_0 in the Ramp diagram.

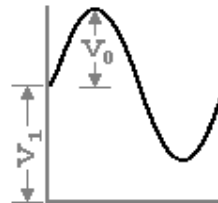
The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:SINusoid:AMPLitude <amplitude>|MIN|MAX, (@<chanlist>)
[SOURCE:]ARB:VOLTage:SINusoid:AMPLitude? [MIN|MAX,](@<chanlist>)

This command specifies the amplitude or peak value of the sine wave in volts. Referenced to V_0 in the Sinusoid diagram.

**Sinusoid
diagram**



The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:SINusoid:FREQuency <frequency>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:SINusoid:FREQuency? [MIN|MAX,](@<chanlist>)

This command specifies the frequency of the sine wave in Hertz. The programming range is from 3.8147E-5 - 1.0E4 Hz.

The *RST value = 1

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:SINusoid:OFFSet <offset>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:SINusoid:OFFSet? [MIN|MAX,](@<chanlist>)

This command specifies the offset of the sine wave from zero, in volts. Because the output cannot generate negative voltages, the offset value cannot be less than the amplitude of the sine wave. Referenced to V_1 in the Sinusoid diagram.

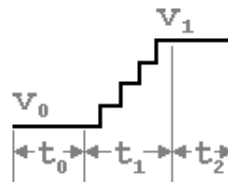
The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:STAircase:END[:LEVel] <voltage>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:STAircase:END[:LEVel]? [MIN|MAX,](@<chanlist>)

This command specifies the end level voltage (after the staircase occurs) in volts. Referenced to V_1 in the Staircase diagram. The voltage difference between the starting voltage and the ending voltage is divided equally between the steps.

**Staircase
diagram**



The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:STAircase:END:TIME <time>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:STAircase:END:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the time in seconds, after the staircase completes, that the end-level voltage persists. End times can be programmed from 0 to 262.143 seconds. Referenced to t_2 in the Staircase diagram.

The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:STAircase:NSteps <time>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:STAircase:NSteps? [MIN|MAX,](@<chanlist>)

This command specifies the number of steps in the staircase. The number of steps that can be programmed range from 0 to 500.

The *RST value = 10

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:STAircase:STArT[:LEVel] <voltage>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:STAircase:STArT[:LEVel]? [MIN|MAX,](@<chanlist>)

This command specifies the initial voltage (before the staircase occurs) in volts. Referenced to V_0 in the Staircase diagram.

The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:STAircase:STArT:TIME <time>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:STAircase:STArT:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the delay in seconds, after the trigger is received, but before the staircase occurs. Start times can be programmed from 0 to 262.143 seconds. Referenced to t_0 in the Staircase diagram.

The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:STAircase:TIME <time>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:STAircase:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the total time to complete all of the staircase steps in seconds. Rise times can be programmed from 0 to 262.143 seconds. Referenced to t_1 in the Staircase diagram.

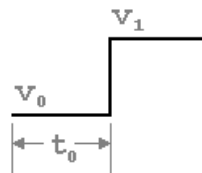
The *RST value = 1

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:STEP:END[:LEVel] <voltage>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:STEP:END[:LEVel]? [MIN|MAX,](@<chanlist>)

This command specifies the step level in volts. Referenced to V_1 in the Step diagram.

**Step
diagram**



The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:STEP:STArT[:LEVel] <voltage>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:STEP:STArT[:LEVel]? [MIN|MAX,](@<chanlist>)

This command specifies the initial voltage level in volts. Referenced to V_1 in the Step diagram.

The *RST value = the minimum programmable value.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:STEP:START:TIME <time>|MIN|MAX, (@<chanlist>)
 [SOURCE:]ARB:VOLTage:STEP:START:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the delay in seconds, after the trigger is received, but before the step occurs. Start times can be programmed from 0 to 262.143 seconds. Referenced to t_0 in the Step diagram.

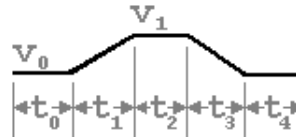
The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:TRAPEZoid:END:TIME <time>|MIN|MAX, (@<chanlist>)
 [SOURCE:]ARB:VOLTage:TRAPEZoid:END:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the time in seconds, after the trapezoid completes, that the start-level voltage persists. End times can be programmed from 0 to 262.143 seconds. Referenced to t_4 in the Trapezoid diagram.

Trapezoid diagram



The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:TRAPEZoid:FTIME <time>|MIN|MAX, (@<chanlist>)
 [SOURCE:]ARB:VOLTage:TRAPEZoid:FTIME? [MIN|MAX,](@<chanlist>)

This command specifies the fall time of the trapezoid in seconds. Fall times can be programmed from 0 to 262.143 seconds. Referenced to t_3 in the Trapezoid diagram.

The *RST value = 1

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:TRAPEZoid:RTIME <time>|MIN|MAX, (@<chanlist>)
 [SOURCE:]ARB:VOLTage:TRAPEZoid:RTIME? [MIN|MAX,](@<chanlist>)

This command specifies the rise time of the trapezoid in seconds. Rise times can be programmed from 0 to 262.143 seconds. Referenced to t_1 in the Trapezoid diagram.

The *RST value = 1

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURCE:]ARB:VOLTage:TRAPEZoid:START[:LEVel] <voltage>|MIN|MAX, (@<chanlist>)
 [SOURCE:]ARB:VOLTage:TRAPEZoid:START[:LEVel]? [MIN|MAX,](@<chanlist>)

This command specifies the initial and final voltage level of the trapezoid in volts. Referenced to V_0 in the Trapezoid diagram.

The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:TRAPezoid:START:TIME <time>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:TRAPezoid:START:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the delay in seconds, after the trigger is received, but before the rising ramp occurs. Start times can be programmed from 0 to 262.143 seconds. Referenced to t_0 in the Trapezoid diagram.

The *RST value = 0

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:TRAPezoid:TOP[:LEVel] <voltage>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:TRAPezoid:TOP[:LEVel]? [MIN|MAX,](@<chanlist>)

This command specifies the top level of the trapezoid in volts. Referenced to V_1 in the Trapezoid diagram.

The *RST value = MIN

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:TRAPezoid:TOP:TIME <time>|MIN|MAX, (@<chanlist>)
[SOURce:]ARB:VOLTage:TRAPezoid:TOP:TIME? [MIN|MAX,](@<chanlist>)

This command specifies the time of the top level of the trapezoid in seconds. Top times can be programmed from 0 to 262.143 seconds. Referenced to t_2 in the Trapezoid diagram.

The *RST value = 1

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:UDEFined:BOSTep[:DATA] ON|OFF {,ON|OFF}, (@<chanlist>)
[SOURce:]ARB:VOLTage:UDEFined:BOSTep[:DATA]? (@<chanlist>)

This command specifies which user-defined ARB points or steps will generate a trigger out signal at the beginning of the step (BOSTep). A comma-delimited list of up to 512 steps may be programmed. The state is either ON (1) or OFF (0). A trigger is only generated when the state is set to ON.

The *RST value = 1 step with a value of OFF.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:UDEFined:BOSTep:POINTs? (@<chanlist>)

This query returns the number of beginning-of-step points in the user-defined ARB, not the point values.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:UDEFined:DWELL <time> {,<time>}, (@<chanlist>)
[SOURce:]ARB:VOLTage:UDEFined:DWELL? (@<chanlist>)

This command specifies the dwell time for each user-defined ARB point. A comma-delimited list of up to 512 points may be programmed. Dwell time is the time that the output will remain at a specific point. Dwell times can be programmed from 0 to 262.143 seconds with the following resolution:

Range	Resolution	Range	Resolution
0 to 0.262143	1 microsecond	0 to 26.2143	100 microseconds
0 to 2.62143	10 microseconds	0 to 262.143	1 millisecond

At the end of the dwell time, the output state of the unit depends upon the ARB:TERM:LAST program settings. The order in which the dwell values are entered determines the sequence of execution.

The *RST value = 1 point with a value of 0.001.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:UDEFined:DWELL:POINTS? (@<chanlist>)

This query returns the number of dwell points in the user-defined ARB, not the point values.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:UDEFined:LEVEL <current> {,<current>}, (@<chanlist>)
[SOURce:]ARB:VOLTage:UDEFined:LEVEL? (@<chanlist>)

This command specifies the voltage setting for each user-defined ARB point in volts. A comma-delimited list of up to 512 points may be programmed.

The *RST value = 1 point with a value of MIN.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]ARB:VOLTage:UDEFined:LEVEL:POINTS? (@<chanlist>)

This query returns the number of voltage points in the user-defined ARB, not the point values.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later.

[SOURce:]CURRENT[:LEVEL][:IMMEDIATE][:AMPLITUDE]<value>|MIN|MAX, (@<chanlist>)
[SOURce:]CURRENT[:LEVEL][:IMMEDIATE][:AMPLITUDE]? [MIN|MAX,] (@<chanlist>)
[SOURce:]CURRENT[:LEVEL]:TRIGGERed[:AMPLITUDE] <value>|MIN|MAX, (@<chanlist>)
[SOURce:]CURRENT[:LEVEL]:TRIGGERed[:AMPLITUDE]? [MIN|MAX,] (@<chanlist>)

These commands set the immediate and the triggered current level of the output channel. The values are programmed in amperes. The immediate level is the output current setting. The triggered level is a stored value that is transferred to the output when a Step transient is triggered. This command is coupled with [SOURce:]CURRENT:RANGE.

The *RST value = 80 mA

[SOURCE:]CURRENT:MODE FIXEd|STEP|LIST|ARB, (@<chanlist>)
[SOURCE:]CURRENT:MODE? (@<chanlist>)

This command determines what happens to the output current when the transient system is initiated and triggered.

FIXed The output voltage remains at the immediate value.

STEP The output goes to the triggered level when a trigger occurs.

LIST The output follows the programmed step value when a trigger occurs.
 This function does not apply to all models (refer to chapter 1 in the User's Guide under "Model Differences").

ARB The output follows the ARBitrary waveform values when a trigger occurs.

The *RST value = FIXed.

[SOURCE:]CURRENT:PROTEction:STATE ON|OFF, (@<chanlist>)
[SOURCE:]CURRENT:PROTEction:STATE? (@<chanlist>)

This command enables or disables the over-current protection (OCP) function. The enabled state is On (1); the disabled state is Off (0). If the over-current protection function is enabled and the output goes into constant current operation, the output is disabled and the Questionable Condition status register OCP bit is set.

The current limit setting determines when the output channel goes into constant current operation. An over-current condition can be cleared with OUTPut:PROTEction:CLEAr after the cause of the condition is removed.

The *RST value = OFF.

[SOURCE:]CURRENT:PROTEction:DELAy[:TIME] <time>|MIN|MAX, (@<chanlist>)
[SOURCE:]CURRENT:PROTEction:DELAy[:TIME]? [MIN|MAX,] (@<chanlist>)

This command sets the over-current protection delay. This prevents momentary changes in output status from triggering the over-current protection function. Programmed values can range from 0 to 255 milliseconds.

The operation of over-current protection is affected by the setting of the current protection delay start event, which is specified by [SOURCE:]CURRENT:PROTEction:DELAy:START.

The *RST value = 20 ms.

Only applies to Agilent N6705A with firmware rev. A.02.00 or later and on N6700 mainframes with firmware rev. C.03.00 or later.

This command is the same as OUTPut:PROTEction:DELAy.

[SOURce:]CURRENT:PROtection:DElay:START SChange|CCTrans, (@<chanlist>)
[SOURce:]CURRENT:PROtection:DElay:START? (@<chanlist>)

This command sets the over-current protection delay start event.

SChange The over-current protection delay timer is started by a programmed *settings change* in the voltage, current, or output state. This is how the OUTPUT:PROtection:DElay command functioned prior to the availability of this command.

CCTrans The over-current protection delay timer is started by any output *transition* to constant current mode.

The *RST value = SCH

Only applies to Agilent N6705A with firmware rev. A.02.00 or later and on N6700 mainframes with firmware rev. C.03.00 or later.

[SOURce:]CURRENT:RANGE <value>|MIN|MAX, (@<chanlist>)
[SOURce:]CURRENT:RANGE? [MIN|MAX,] (@<chanlist>)

This command sets the output current range. Units are in amperes. The instrument selects the range with the best resolution for the value entered. When queried, the returned value is the maximum DC current that can be output on the range that is presently set.

This command is coupled with the [SOURce:]CURRENT command. This means that if a range command is sent that places an output on a range with a lower maximum current than the present current level, an error is generated. This also occurs if a current level is programmed with a value too large for the present range.

These types of errors can be avoided by sending the both level and range commands in the same SCPI message. When the range and setting information is received as a set, no range/setting conflict occurs.

The *RST value = the highest available range.

NOTE

If programming a range value causes a range change to occur while the output is enabled, the output will be temporarily disabled while the range switch occurs. The transition from on-to-off and then from off-to-on will also be delayed by the settings of OUTPUT:DElay:FALL and OUTPUT:DElay:RISE.

This command only applies to models that have programmable ranges. Refer to Appendix A in the User's Guide for the available ranges for each model.

[SOURce:]DIGital:INPut:DATA?

This query reads the state of the digital control port. The query returns the state of pins 1 through 7 in bits 0 through 6 respectively.

[SOURCE:]DIGital:OUTPut:DATA <value>

[SOURCE:]DIGital:OUTPut:DATA?

This command sets the output data on the digital control port when that port is configured for Digital I/O operation. The port has seven signal pins and a digital ground pin. In the binary-weighted value that is written to the port, the pins are controlled according to the following bit assignments.

Pin	1	2	3	4	5	6	7
Bit	0	1	2	3	4	5	6
Decimal Value	1	2	4	8	16	32	64

The query returns the last programmed value of the bits. To read the actual state of the pin, use [SOURCE:]DIGital:INPut:DATA?

[SOURCE:]DIGital:PIN1:FUNCTION DIO|DINPut|TOUTput|TINPut|FAULT

[SOURCE:]DIGital:PIN2:FUNCTION DIO|DINPut|TOUTput|TINPut

[SOURCE:]DIGital:PIN3:FUNCTION DIO|DINPut|TOUTput|TINPut|INHibit

[SOURCE:]DIGital:PIN4:FUNCTION DIO|DINPut|TOUTput|TINPut|ONCouple|OFFCouple

[SOURCE:]DIGital:PIN5:FUNCTION DIO|DINPut|TOUTput|TINPut|ONCouple|OFFCouple

[SOURCE:]DIGital:PIN6:FUNCTION DIO|DINPut|TOUTput|TINPut|ONCouple|OFFCouple

[SOURCE:]DIGital:PIN7:FUNCTION DIO|DINPut|TOUTput|TINPut|ONCouple|OFFCouple

[SOURCE:]DIGital:PIN<1-7>:FUNCTION?

These commands set the functions of the digital port pins. The pin functions are saved in non-volatile memory.

DIO The pin is a general-purpose ground-referenced digital input/output. The output can be set with [SOURCE:]DIGital:OUTPut:DATA <value>.

DINPut The pin is in digital input-only mode. The digital output data of the corresponding pin is ignored.

TOUTput The pin is configured as a trigger output. When configured as a trigger output, the pin will only generate output triggers if the Step or List transient system has been configured to generated trigger signals.

TINPut The pin is configured as a trigger input. When configured as a trigger input, the pin can be selected as a source of measurement and transient trigger signals.

FAULT Applies only to pin 1. Setting FAULT means that pin 1 functions as an isolated fault output. The fault signal is true when any output is in a protected state (from OCP, OVP, OT, PF, or INH). Note also that Pin 2 serves as the isolated common for pin 1. When pin 1 is set to the FAULT function, the instrument ignores any commands to program pin 2. Queries of pin 2 will return FAULT. If pin 1 is changed from FAULT to another function, pin 2 is set to DINPut.

INHibit Applies only to pin 3. When pin 3 is configured as an inhibit input, a true signal at the pin will disable all output channels.

ONCouple Applies only to pins 4 -7. When configured as an On control, the pin will synchronize the output On state between mainframes. Only one pin can be configured as an On control. The pin will function as both an input and an output. The polarity of the pin is fixed and cannot be programmed.

OFFCouple Applies only to pins 4-7. When configured as an Off control, the pin will synchronize the output Off state between mainframes. Only one pin can be configured as an Off control. The pin will function as both an input and an output. The polarity of the pin is fixed and cannot be programmed.

[SOURCE:]DIGital:PIN<1-7>:POLarity POSitive|NEGative
[SOURCE:]DIGital:PIN<1-7>:POLarity?

These commands set the polarity of the digital port pins. The pin polarities are saved in non-volatile memory.

Setting a polarity to POSitive means that a logical true signal is a voltage high at the pin. Setting the polarity NEGative means that a logical true signal is a voltage low at the pin. For trigger inputs and outputs, POSitive means a rising edge; NEGative means a falling edge.

[SOURCE:]LIST:COUNT <count>|MIN|MAX|INF, (@<chanlist>)
[SOURCE:]LIST:COUNT? [MIN|MAX,] (@<chanlist>)

This command sets the number of times that the list is executed before it is completed. Applies only to models with list capability. The range is 1 through 256. Use INFinity to execute a list indefinitely. In this case, use ABORt:TRANsient to stop the list.

The *RST value = 1

[SOURCE:]LIST:CURRENT[:LEVel] <curr> {,<curr>}, (@<chanlist>)
[SOURCE:]LIST:CURRENT[:LEVel]? (@<chanlist>)

This command specifies the current setting for each list step in amperes. Applies only to models with list capability. A comma-delimited list of up to 512 steps may be programmed.

The *RST value = 1 step with a value of MIN.

[SOURCE:]LIST:CURRENT:POINTS? (@<chanlist>)

This query returns the number of points (steps) programmed in the current list. Applies only to models with list capability.

[SOURCE:]LIST:DWELL <time> {,<time>}, (@<chanlist>)
[SOURCE:]LIST:DWELL? (@<chanlist>)

This command specifies the dwell time for each list step. Applies only to models with list capability. A comma-delimited list of up to 512 steps may be programmed. Dwell time is the time that the output will remain at a specific step. Dwell times can be programmed from 0 to 262.143 seconds with the following resolution:

Range	Resolution	Range	Resolution
0 to 0.262143	1 microsecond	0 to 26.2143	100 microseconds
0 to 2.62143	10 microseconds	0 to 262.143	1 millisecond

At the end of the dwell time, the output state of the unit depends upon the LIST:STEP program settings. See LIST:STEP

The order in which the values are entered determines the sequence when the list executes.

The *RST value = 1 step with a value of 0.001.

[SOURCE:]LIST:DWELL:POINTS? (@<chanlist>)

This query returns the number of points (steps) in the dwell list.
Applies only to models with list capability.

[SOURCE:]LIST:STEP ONCE|AUTO, (@<chanlist>)**[SOURCE:]LIST:STEP? (@<chanlist>)**

This command specifies how the list responds to triggers. Applies only to models with list capability.

ONCE Causes the output to remain at the present step until a trigger advances it to the next step. Triggers that arrive during the dwell time are ignored.

AUTO Causes the output to automatically advance to each step, after the receipt of an initial starting trigger. The steps are paced by the dwell list. As each dwell time elapses, the next step is immediately output.

The *RST value = AUTO.

[SOURCE:]LIST:TERMinate:LAST ON|OFF, (@<chanlist>)**[SOURCE:]LIST:TERMinate:LAST? (@<chanlist>)**

This command determines the output value when the list terminates. Applies only to models with list capability. The state is either ON (1) or OFF (0). When ON, the output voltage or current remains at the value of the last list step. The value of the last voltage or current list step becomes the IMMEDIATE value when the list completes. When OFF, and also when the list is aborted, the output returns to the settings it was at before the list started.

The *RST value = OFF.

[SOURCE:]LIST:TOUTput:BOSTep[:DATA] ON|OFF {,ON|OFF}, (@<chanlist>)**[SOURCE:]LIST:TOUTput:BOSTep[:DATA]? (@<chanlist>)**

This command specifies which list steps generate a trigger out signal at the beginning of the list step (BOSTep). Applies only to models with list capability. A comma-delimited list of up to 512 steps may be programmed. The state is either ON (1) or OFF (0). A trigger is only generated when the state is set to ON.

The *RST value = 1 step with a value of OFF.

[SOURCE:]LIST:TOUTput:EOSTep[:DATA] ON|OFF {,ON|OFF}, (@<chanlist>)**[SOURCE:]LIST:TOUTput:EOSTep[:DATA]? (@<chanlist>)**

This command specifies which list steps generate a trigger out signal at the end of the list step's (EOSTep) dwell time. Applies only to models with list capability. A comma-delimited list of up to 512 steps may be programmed. The state is either ON (1) or OFF (0). A trigger is only generated when the state is set to ON.

The *RST value = 1 step with a value of OFF.

[SOURCE:]LIST:VOLTage[:LEVel] <volt> {,<volt>}, (@<chanlist>)
[SOURCE:]LIST:VOLTage[:LEVel]? (@<chanlist>)

This command specifies the voltage setting for each list step in volts. Applies only to models with list capability. Up to 512 steps may be programmed. The values are separated by commas.

The *RST value = 1 step with a value of MIN.

[SOURCE:]LIST:VOLTage:POINTs? (@<chanlist>)

This query returns the number of points (steps) in the voltage list, not the point values. Applies only to models with list capability.

[SOURCE:]POWer:LIMit <power>|MIN|MAX, (@<chanlist>)
[SOURCE:]POWer:LIMit? [MIN|MAX,] (@<chanlist>)

This command sets the power limit on output channels. Units are in Watts.

For the majority of Agilent N6700 and Agilent N6705A instrument configurations, full power is available from all installed power modules. However, it is possible to configure an instrument in which the combined ratings of the power modules exceed the power rating of the mainframe. For these instruments, power allocation lets you programmatically limit the output power of some of the power modules so that the combined output power does not exceed the power rating of the mainframe. Refer to Appendix C in the User's Guide for more information.

Setting a power limit of MAX on an output channel sets the output channel to its maximum rated power limit. If the power limit is left at the maximum rating, then the power module will not enable its power limit function.

Querying MAX on an output channel returns the maximum rated power limit.

The *RST value = MAX.

NOTE

Power limiting should not be used when Agilent N673xB, N674xB, and N677xA power modules are grouped. When these power modules are grouped, you should reset the power limit to its maximum rated value.

[SOURCE:]STEP:TOUTput ON|OFF, (@<chanlist>)
[SOURCE:]STEP:TOUTput? (@<chanlist>)

This command specifies whether an output trigger signal is generated when a transient voltage or current step occurs. The state is either ON (1) or OFF (0). A trigger is generated when the state is True.

The *RST value = OFF.

```
[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude]<value>|MIN|MAX, (@<chanlist>)
[SOURce:]VOLTage[:LEVel][:IMMediate][:AMPLitude]? [MIN|MAX,] (@<chanlist>)
[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude] <value>|MIN|MAX, (@<chanlist>)
[SOURce:]VOLTage[:LEVel]:TRIGgered[:AMPLitude]? [MIN|MAX,] (@<chanlist>)
```

These commands set the immediate and the triggered voltage level of the output channel. The values are programmed in volts. The immediate level is the output voltage setting. The triggered level is a stored value that is transferred to the output when a Step transient is triggered. This command is coupled with [SOURce:]VOLTage:RANGE.

The *RST value = MIN.

```
[SOURce:]VOLTage:MODE FIXEd|STEP|LIST|ARB, (@<chanlist>)
[SOURce:]VOLTage:MODE? (@<chanlist>)
```

This command determines what happens to the output voltage when the transient system is initiated and triggered.

FIXEd The output voltage remains at the immediate value.

STEP The output goes to the triggered level when a trigger occurs.

LIST The output follows the programmed list step value when a trigger occurs. This function does not apply to all models (refer to chapter 1 in the User's Guide under "Model Differences").

ARB The output follows the ARBitrary waveform values when a trigger occurs.

The *RST value = FIXEd.

```
[SOURce:]VOLTage:PROTection[:LEVel] <value>|MIN|MAX, (@<chanlist>)
[SOURce:]VOLTage:PROTection[:LEVel]? [MIN|MAX,] (@<chanlist>)
```

This command sets the over-voltage protection (OVP) level of the output channel. The values are programmed in volts. If the output voltage exceeds the OVP level, the output is disabled and the Questionable Condition status register OV bit is set. An over-voltage condition can be cleared with the Output Protection Clear command after the condition that caused the OVP trip is removed.

The *RST value = MAX.

[SOURce:]VOLTage:RANGe <value>|MIN|MAX, (@<chanlist>)
[SOURce:]VOLTage:RANGe? [MIN|MAX,] (@<chanlist>)

This command sets the output voltage range. Units are in volts. The instrument selects the range with the best resolution for the value that is entered. When queried, the returned value is the maximum voltage that can be output on the range that is presently set.

This command is coupled with the [SOURce:]VOLTage command. This means that if a range command is sent that places an output on range with a lower maximum voltage than the present voltage level, an error is generated. This also occurs if a voltage level is programmed with a value too large for the present range.

These types of errors can be avoided by sending the both level and range commands in the same SCPI message. When range and setting information is received as a set, no range/setting conflict occurs.

The *RST value = the highest available range.

NOTE

If programming a range value causes a range change to occur while the output is enabled, the output will be temporarily disabled while the range switch occurs. The transition from on-to-off and then from off-to-on will also be delayed by the settings of OUTPUT:DElay:FALL and OUTPUT:DElay:RISE.

This command only applies to models that have programmable ranges. Refer to Appendix A in the User's Guide for the available ranges for each model.

[SOURce:]VOLTage:SENse:SOURce ON|OFF, (@<chanlist>)
[SOURce:]VOLTage:SENse:SOURce? (@<chanlist>)

This command sets the state of the remote sense relays.

Internal Internal sets the remote sense relays to local sensing. The front panel remote sense terminals are internally connected to the output terminals. The 4 wire indicator is off.

External External sets the remote sense relays to remote sensing. The front panel remote sense terminals are enabled and must be connected to the external load. The 4 wire indicator is on.

Only applies to Agilent N6705A.

```
[SOURce:]VOLTage:SLEW[:IMMediate] <value>|MIN|MAX|INF, (@<chanlist>)
[SOURce:]VOLTage:SLEW[:IMMediate]? [MIN|MAX,] (@<chanlist>)
```

This command sets the voltage slew rate in volts per second. The slew rate setting affects all programmed voltage changes, including those due to the output state turning on or off. The slew rate can be set to any value between 0 and 9.9E+37. For very large values, the slew rate will be limited by the analog performance of the output circuit. The keywords MAXimum or INFinity set the slew rate to maximum.

Internally, the slew rate is controlled by a 24-bit register. The slowest or minimum slew rate is a function of the full-scale voltage range. For a model with a 50 V range, the minimum slew rate is about 4.76 V/s. For other voltage ranges the minimum slew rate is proportional to this value, so for a model with a 5 V range the minimum slew rate is about 0.476 V/s. The unit accepts slew rates as low as 0 V/s, but values sent to the 24-bit register will be limited at 1 count.

The query returns the value that was sent, unless the value was less than the minimum slew rate, in which case the minimum value is returned. The LSB weight of the 24-bit register can be queried using VOLT:SLEW? MIN. The exact value varies slightly according to the voltage calibration.

The *RST value = 9.9E+37.

Status Subsystem

Status register programming lets you determine the operating condition of the instrument at any time. The instrument has three groups of status registers; Operation, Questionable, and Standard Event. The Operation and Questionable status groups each consist of Condition, Enable, and Event registers and NTR and PTR filters.

The Status subsystem is also programmed using Common commands. Common commands control additional status functions such as the Service Request Enable and the Status Byte registers.

Operation Status Group

These registers record signals that occur during normal operation. The group consists of a Condition, PTR/NTR, Event, and Enable register. The outputs of the Operation Status register group are logically-ORed into the OPERation summary bit (7) of the Status Byte register.

Questionable Status Group

These registers record signals that indicate abnormal operation. The group consists of the same register types as the Operation Status group. The outputs of the Questionable Status group are logically-ORed into the QUEStionable summary bit (3) of the Status Byte register.

Standard Event Status Group

These registers are programmed by Common commands. The Standard Event event register latches events relating to communication status. It is a read-only register that is cleared when read. The Standard Event enable register functions similarly to the enable registers of the Operation and Questionable status groups.

Status Byte Register

This register summarizes the information from all other status groups as defined in the *IEEE 488.2 Standard Digital Interface for Programmable Instrumentation*.

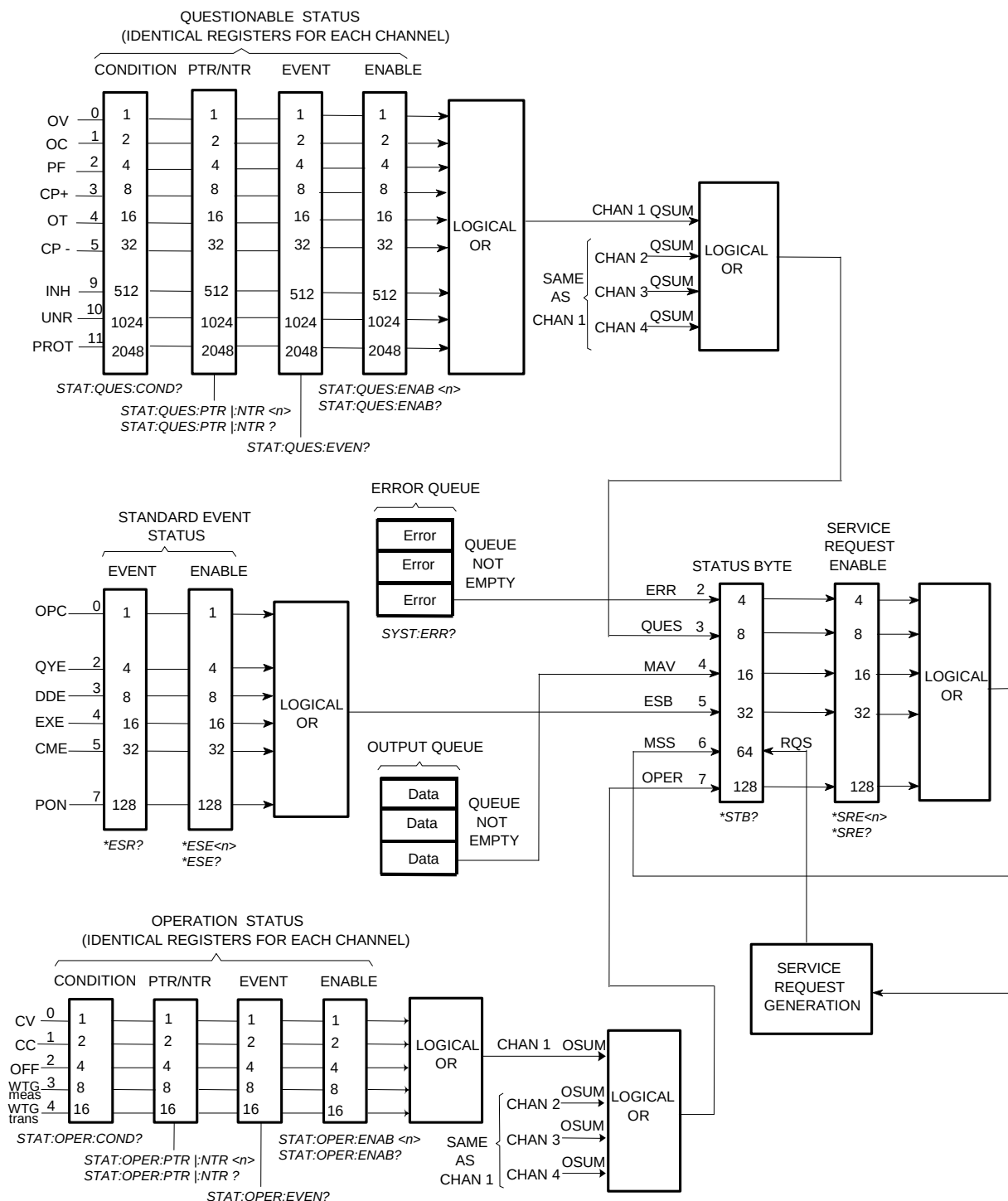
MSS and RQS Bits

MSS is a real-time (unlatched) summary of all Status Byte register bits that are enabled by the Service Request Enable register. MSS is set whenever the instrument has one or more reasons for requesting service. *STB? reads the MSS in bit position 6 of the response but does not clear any of the bits in the Status Byte register.

The RQS bit is a latched version of the MSS bit. Whenever the instrument requests service, it sets the SRQ interrupt line true and latches RQS into bit 6 of the Status Byte register. When the controller does a serial poll, RQS is cleared inside the register and returned in bit position 6 of the response. The remaining bits of the Status Byte register are not disturbed.

MAV Bit and Output Queue

The Output Queue is a first-in, first-out (FIFO) data register that stores instrument-to-controller messages until the controller reads them. Whenever the queue holds one or more bytes, it sets the MAV bit (4) of the Status Byte register.



STATus:PRESet

This command sets all defined bits in the Status system's PTR registers and clears all bits in the NTR and Enable registers.

Operation Register	Questionable Register	Preset Settings
STAT:OPER:ENAB	STAT:QUES:ENAB	0 all bits disabled
STAT:OPER:NTR	STAT:QUES:NTR	0 all bits disabled
STAT:OPER:PTR		31 all defined bits enabled
	STAT:QUES:PTR	3647 all defined bits enabled

STATus:OPERation[:EVENT]? (@<chanlist>)

This query returns the value of the Operation Event register. The Event register is a read-only register, which stores (latches) all events that are passed by the Operation NTR and/or PTR filter. Reading the Operation Event register clears it. The bit configuration of the Operation status registers is as follows:

Bit Position	15-5	4	3	2	1	0
Bit Value	—	16	8	4	2	1
Bit Name	—	WTG-tran	WTG-meas	OFF	CC	CV

WTG-tran = The transient system is waiting for a trigger.

WTG-meas = The measurement system is waiting for a trigger

OFF = The output is programmed off

CC = The output is in constant current

CV = The output is in constant voltage

STATus:OPERation:CONDition? (@<chanlist>)

This query returns the value of the Operation Condition register. That is a read-only register, which holds the live (unlatched) operational status of the instrument.

STATus:OPERation:ENABLE <value>, (@<chanlist>)**STATus:OPERation:ENABLE? (@<chanlist>)**

This command and its query set and read the value of the Operational Enable register. This register is a mask for enabling specific bits from the Operation Event register to set the operation summary bit (OPER) of the Status Byte register. This bit (bit 7) is the logical OR of all the Operational Event register bits that are enabled by the Status Operation Enable register.

STATus:OPERation:NTRansition <value>, (@<chanlist>)
STATus:OPERation:PTRansition <value>, (@<chanlist>)
STATus:OPERation:NTRansition? (@<chanlist>)
STATus:OPERation:PTRansition? (@<chanlist>)

These commands set and read the value of the Operation NTR (Negative-Transition) and PTR (Positive-Transition) registers. These registers serve as polarity filters between the Operation Condition and Operation Event registers to cause the following actions:

- When a bit in the Operation NTR register is set to 1, then a 1-to-0 transition of the corresponding bit in the Operation Condition register causes that bit in the Operation Event register to be set.
- When a bit of the Operation PTR register is set to 1, then a 0-to-1 transition of the corresponding bit in the Operation Condition register causes that bit in the Operation Event register to be set.
- If the same bits in both NTR and PTR registers are set to 1, then any transition of that bit at the Operation Condition register sets the corresponding bit in the Operation Event register.
- If the same bits in both NTR and PTR registers are set to 0, then no transition of that bit at the Operation Condition register can set the corresponding bit in the Operation Event register.

STATus:QUESTionable[:EVENT]? (@<chanlist>)

This query returns the value of the Questionable Event register. The Event register is a read-only register, which stores (latches) all events that are passed by the Questionable NTR and/or PTR filter. Reading the Questionable Event register clears it. The bit configuration of the Questionable status registers is as follows:

Bit Position	15-12	11	10	9	8-6	5	4	3	2	1	0
Bit Value	–	2048	1024	512	–	32	16	8	4	2	1
Bit Name	–	PROT	UNR	INH	–	CP –	OT	CP+	PF	OC	OV

PROT = The output has been disabled because it is coupled to a protection condition that occurred on another channel.

UNR = The output is unregulated

INH = The output is inhibited by an external signal

CP– = The output is limited by the negative power limit

OT = The over-temperature protection has tripped

CP+ = The output is limited by the positive power limit

PF = The output is disabled by the power-fail - which may be caused by a low-line or brownout condition on the AC line

OC = The output is disabled by the over-current protection

OV = The output is disabled by the over-voltage protection

STATus:QUESTionable:CONDition? (@<chanlist>)

This query returns the value of the Questionable Condition register. That is a read-only register, which holds the real-time (unlatched) questionable status of the instrument.

STATUS:QUESTIONable:ENABLE <value>, (@<chanlist>)
STATUS:QUESTIONable:ENABLE? (@<chanlist>)

This command and its query set and read the value of the Questionable Enable register. This register is a mask for enabling specific bits from the Questionable Event register to set the questionable summary bit (QUES) of the Status Byte register. This bit (bit 3) is the logical OR of all the Questionable Event register bits that are enabled by the Questionable Status Enable register.

STATUS:QUESTIONable:NTRansiton <value>, (@<chanlist>)
STATUS:QUESTIONable:PTRansiton <value>, (@<chanlist>)
STATUS:QUESTIONable:NTRansition? (@<chanlist>)
STATUS:QUESTIONable:PTRansition? (@<chanlist>)

These commands set or read the value of the Questionable NTR (Negative-Transition) and PTR (Positive-Transition) registers. These registers serve as polarity filters between the Questionable Condition and Questionable Event registers to cause the following actions:

- When a bit of the Questionable NTR register is set to 1, then a 1-to-0 transition of the corresponding bit of the Questionable Condition register causes that bit in the Questionable Event register to be set.
- When a bit of the Questionable PTR register is set to 1, then a 0-to-1 transition of the corresponding bit in the Questionable Condition register causes that bit in the Questionable Event register to be set.
- If the same bits in both NTR and PTR registers are set to 1, then any transition of that bit at the Questionable Condition register sets the corresponding bit in the Questionable Event register.
- If the same bits in both NTR and PTR registers are set to 0, then no transition of that bit at the Questionable Condition register can set the corresponding bit in the Questionable Event register.

System Commands

System commands control system functions that are not directly related to output control, measurement, or status functions. Common commands are also used to control system functions.

SYSTem:CHANnel[:COUNT]?

This query returns the number of output channels in a mainframe.

SYSTem:CHANnel:MODeL? (@<chanlist>)

This query returns the model numbers of the selected output channels. Model numbers are comma-delimited.

SYSTem:CHANnel:OPTion? (@<chanlist>)

This query returns a list of options installed in each channel given in the channel list. The list of options for each channel is surrounded by double quotes. If there are no options installed in a channel, an empty pair of double quotes is returned.

SYSTem:CHANnel:SERial? (@<chanlist>)

This query returns the serial numbers of the selected output channels. Serial numbers are comma-delimited.

SYSTem:COMMunicate:RLState LOCAL|REMOte|RWLock SYSTem:COMMunicate:RLState?

This command configures the remote/local state of the instrument according to the following settings.

LOCAL The instrument is set to front panel and remote interface control.

REMOte The instrument is set to front panel and remote interface control.

RWLock The front panel keys are disabled. The instrument can only be controlled via the remote interface. This programmable setting is completely independent from the front panel lock/unlock function that is available from the front panel menu.

The remote/local instrument state can also be set by other interface commands over the GPIB and some other I/O interfaces. When multiple remote programming interfaces are active, the interface with the most recently changed remote/local state determines the instrument's remote/local state.

The remote/local state is unaffected by *RST or any SCPI commands other than SYST:COMM:RLST. At power-on, the state is LOCAL.

SYSTem:COMMunicate:TCPip:CONTRol?

This query returns the control connection port number. This is used to open a control socket connection to the instrument. Refer to chapter 3 in the User's Guide under "Using Sockets".

SYSTem:ERRor?

This query returns the next error number and its corresponding message string from the error queue. The queue is a FIFO (first-in, first-out) buffer that stores errors as they occur. As it is read, each error is removed from the queue. When all errors have been read, the query returns 0, "No error". If more errors are accumulated than the queue can hold, the last error in the queue will be -350, "Too many errors" (see appendix A for error codes).

SYSTem:DATE <yyyy>,<mm>,<dd>

This command sets the date of the real time clock in year (yyyy), month (mm), and day (dd). Only applies to Agilent N6705A.

SYSTem:GRoup:CATalog?

This query returns information about channels that are grouped. The defined groups are enclosed in quotes. For example, the returned string "1,2,3", "4" indicates that channels 1, 2, and 3 are grouped. Channel 4 is not grouped, as it appears by itself in the quote string.

Does not apply to Agilent N6705A.

SYSTem:GRoup:DEFine (@<chanlist>)

This command defines a list of output channels as a paralleled group. This effectively creates a single output with higher current and power capability. You can group up to four channels per mainframe. Channels must have identical model numbers and options installed.

After the channels are wired in parallel and defined as a group, they can be addressed using any of the channel-specific SCPI commands by sending the channel number of the **lowest** channel in the group.

Group channel lists are stored in non-volatile memory and are unaffected by *RST or *RCL. But the group channel settings (voltage, current, etc.) **are** set and saved by *RST or *RCL.

This command deletes any previously saved states. However, for the group changes to take effect, you must also reboot the unit. Either cycle AC power or send the SYSTem:REBoot command.

Does not apply to Agilent N6705A.

SYSTem:GRoup:DELeTe <channel>

This command removes the indicated channel from a group. It leaves the other channels in the group intact.. When ungrouping a channel, you must also remove the parallel connections between the output and sense terminals of that channel.

This command deletes any previously saved states. However, for the group changes to take effect, you must also reboot the unit. Either cycle AC power or send the SYSTem:REBoot command.

Does not apply to Agilent N6705A.

SYSTem:GROup:DELeTe:ALL

This command restores a group of channels that have been grouped back to an ungrouped state. When ungrouping channels, you must also remove all paralleled connections between channels.

This command deletes any previously saved states. However, for the group changes to take effect, you must also reboot the unit. Either cycle AC power or send the SYSTem:REBoot command.

Does not apply to Agilent N6705A.

SYSTem:PASSword:FPANel:RESet

This command resets the front panel lockout password to the factory-shipped setting, which is zero (0). This command does not reset the calibration password.

NOTE

The front panel password can also be reset to 0 by setting an internal switch on the unit. This switch will also reset the calibration password to 0. Refer to chapter 3 in the Service Guide under "Calibration Switches" for more information.

SYSTem:REBoot

This command causes the instrument to reboot to its power-on state.

SYSTem:Time <hh>,<mm>,<ss>

This command sets the real time clock in hours (hh), minutes (mm), and seconds (ss). Only applies to Agilent N6705A.

SYSTem:VERSion?

This query returns the SCPI version number to which the instrument complies. The returned value is of the form YYYY.V, where YYYY represents the year and V is the revision number for that year.

Trigger Subsystem

Trigger commands control the remote triggering of the instrument. They specify the trigger source for the transient and the measurement system and also generate software triggers.

TRIGger:ACQuire[:IMMediate] (@<chanlist>)

This command sends an immediate trigger to the measurement system. When the trigger system is initiated, a measurement trigger causes the instrument to measure the output voltage or current and store the results in a buffer. The measured quantity, voltage or current is specified by the SENSE:FUNCTION command.

TRIGger:ACQuire:SOURce BUS|PIN<n>|TRANSient<n>, (@<chanlist>) TRIGger:ACQuire:SOURce? (@<chanlist>)

This command selects the trigger source for the measurement trigger system. The following trigger sources can be selected:

BUS Selects GPIB device trigger, *TRG, or <GET> (Group Execute Trigger).

PIN<n> Selects a pin on the external port connector as the trigger source. <n> specifies the pin number. [SOURce:]DIGital:PIN<n>:FUNCTION programs the function of each pin. [SOURce:]DIGital:PIN<n>:POLarity programs the polarity of each pin.

TRANSient<n> Selects the output channel's transient system as the trigger source. <n> specifies the channel. To generate triggers from the transient system use: [SOURce:]STEP:TOUTput, [SOURce:]LIST:TOUTput:BOSTep, and [SOURce:]LIST:TOUTput:EOSTep.

The *RST value = BUS.

TRIGger:ARB:SOURce IMMediate|EXternal|BUS TRIGger:ARB:SOURce?

This command selects the trigger source for the arbitrary waveforms. The following trigger sources can be selected:

IMMediate Triggers the arbitrary waveform immediately.

EXternal Selects the rear panel Trigger In connector.

BUS Selects a remote interface trigger command.

Only applies to Agilent N6705A.

TRIGger:DLOG[:IMMediate]

This command sends an immediate trigger signal to the data logger. This will trigger the data log regardless of the selected data logger trigger source. You must initiate (run) the data logger before you trigger it.

Only applies to Agilent N6705A.

TRIGger:DLOG:CURRENT[:LEVel] <value>|MIN|MAX (@<chanlist>)
TRIGger:DLOG:CURRENT[:LEVel]? [MIN|MAX,] (@<chanlist>)

This command sets the triggered current level of the specified output channel when the trigger source for the data logger is set to current. Units are in amperes. The output channel specified with this command must match the output channel specified by TRIGger:DLOG:SOURce.

Only applies to Agilent N6705A.

TRIGger:DLOG:CURRENT:SLOPe POSitive|NEGative, (@<chanlist>)
TRIGger:DLOG:CURRENT:SLOPe? (@<chanlist>)

This command sets the slope of the current level of the specified output channel when the trigger source for the data logger is set to current. The output channel specified with this command must match the output channel specified by TRIGger:DLOG:SOURce.

POSitive Specifies a rising slope of the output signal.

NEGative Specifies a falling slope of the output signal.

Only applies to Agilent N6705A.

TRIGger:DLOG:SOURce IMMEDIATE|EXTERNAL|BUS|VOLTage<n>|CURRENT<n>|ARSkey|000Key
TRIGger:DLOG:SOURce? (@<chanlist>)

This command selects the trigger source for the data logger. The following trigger sources can be selected:

IMMEDIATE Triggers the data logger immediately.

EXTERNAL Selects the rear panel Trigger In connector.

BUS Selects a remote interface trigger command.

VOLTage<n> Selects a voltage level. <n> specifies the output channel.

CURRENT<n> Selects a current level. <n> specifies the output channel.

ARSkey Selects the Arb Run/Stop key.

000Key Selects the Output On/Off key or the All Outputs On/Off key.

Only applies to Agilent N6705A.

TRIGger:DLOG:VOLTage[:LEVel] <value>|MIN|MAX (@<chanlist>)
TRIGger:DLOG:VOLTage[:LEVel]? [MIN|MAX,] (@<chanlist>)

This command sets the triggered voltage level of the specified output channel when the trigger source for the data logger is set to voltage. Units are in volts. The output channel specified with this command must match the output channel specified by TRIGger:DLOG:SOURce.

Only applies to Agilent N6705A.

TRIGger:DLOG:VOLTage:SLOPe POSitive|NEGative, (@<chanlist>)
TRIGger:DLOG:VOLTage:SLOPe? (@<chanlist>)

This command sets the slope of the voltage level of the specified output channel when the trigger source for the data logger is set to voltage. The output channel specified with this command must match the output channel specified by TRIGger:DLOG:SOURce.

POSitive Specifies a rising slope of the output signal.

NEGative Specifies a falling slope of the output signal.

Only applies to Agilent N6705A.

TRIGger:TRANSient[:IMMediate] (@<chanlist>)

This command generates an immediate transient trigger regardless of the selected trigger source. Output triggers affect the following functions: voltage, current, and current limit. You must initiate the output trigger system before you can send any triggers. When sent, the output trigger will:

- Initiate an output change as specified by the Current Triggered or Voltage Triggered commands.
- Clears the WTG-tran bit in the Status Operation Condition register after the transient trigger sequence has completed.

TRIGger:TRANSient:SOURce BUS|PIN<n>|TRANSient<n>, (@<chanlist>)
TRIGger:TRANSient:SOURce? (@<chanlist>)

This command selects the trigger source for the output trigger system. The following trigger sources can be selected:

BUS Selects a remote interface trigger command.

PIN<n> Selects a pin on the external port connector as the trigger source. <n> specifies the pin number. [SOURce:]DIGital:PIN<n>:FUNctIon programs the function of each pin. [SOURce:]DIGital:PIN<n>:POLarity programs the polarity of each pin.

TRANSient<n> Selects the output channel's transient system as the trigger source. <n> specifies the channel. To generate triggers from the transient system use: [SOURce:]STEP:TOUTput, [SOURce:]LIST:TOUTput:BOSTep, and [SOURce:]LIST:TOUTput:EOSTep.

The *RST value = BUS.

Common Commands

Common commands generally control overall power system functions, such as reset, status, and synchronization. All common commands consist of a three-letter mnemonic preceded by an asterisk: *RST *IDN? *SRE 8

*CLS

This command causes the following actions on the status system:

- Clears the Standard Event Status, Operation Status Event, and Questionable Status Event registers
- Clears the Status Byte and the Error Queue
- If *CLS immediately follows a program message terminator (<NL>), then the output queue and the MAV bit are also cleared.

*ESE

*ESE?

This command programs the Standard Event Status Enable register bits. The programming determines which events of the Standard Event Status Event register (see *ESR?) are allowed to set the ESB (Event Summary Bit) of the Status Byte register. A "1" in the bit position enables the corresponding event.

All of the enabled events of the Standard Event Status Event Register are logically ORed to cause the Event Summary Bit (ESB) of the Status Byte Register to be set. The query reads the Standard Event Status Enable register. The bit configuration of the Standard Event register is as follows:

Bit Position	7	6	5	4	3	2	1	0
Bit Value	128	–	32	16	8	4	–	1
Bit Name	PON	–	CME	EXE	DDE	QUE	–	OPC

PON = Power-on has occurred

CME = Command error

EXE = Execution error

DDE = Device-dependent error

QUE = Query error

OPC = Operation complete

*ESR?

This query reads the Standard Event Status Event register. Reading the register clears it. The bit configuration is the same as the Standard Event Status Enable register (see *ESE).

***IDN?**

This query requests the instrument to identify itself. It returns a string of four fields separated by commas.

Field	Information
Agilent Technologies	Manufacturer
N67xxA	Mainframe model number followed by a letter suffix
0	Zero or mainframe serial number if available
<A>.xx.xx	Revision levels of firmware

OPC**OPC?**

The command is mainly used for program synchronization. It causes the instrument to set the OPC bit (bit 0) of the Standard Event Status register when the instrument has completed all pending operations sent before the *OPC command. *Pending operations* are complete when:

- All commands sent before *OPC, including paralleled commands, have been completed. Most commands are sequential and are completed before the next command is executed. Commands that affect output voltage, current, or state, relays, and trigger actions are executed in parallel with subsequent commands. *OPC provides notification that all parallel commands have completed.
- All triggered actions are completed

*OPC does not prevent processing of subsequent commands, but the OPC bit will not be set until all pending operations are completed.

*OPC? causes the instrument to place an ASCII "1" in the Output Queue when all pending operations are completed. *OPC? does not suspend processing of commands.

***OPT?**

This query requests the mainframe to identify any installed options. A 0 indicates no options are installed.

***RCL <state>**

This command restores the instrument to a state that was previously stored in memory locations 0 through 1 with the *SAV command. All instrument states are recalled except for the following:

- The trigger system is set to the Idle state by an implied ABORT command (this cancels any uncompleted trigger actions).
- Calibration is disabled by setting CALibration:STaTe to OFF.
- All list settings are set to their *RST values.

NOTE

The device state stored in location 0 is automatically recalled at power turn-on when the Output Power-On state is set to RCL 0.

***RDT?**

This query returns a description of the output channels installed in a mainframe. Semicolons separate multiple channel descriptions.

Field	Information
CHAN <c>	Channel number
description	Description of the output channel

***RST**

This command resets the volatile memory of the instrument to a factory-defined state (see “*RST Settings” at the beginning of this chapter).

*RST also forces the ABORT:ACQuire and ABORT:TRANsient commands. This cancels any measurement or output trigger actions presently in process, and resets the two WTG bits in the Status Operation Condition register.

SAV <state>*CAUTION**

This command causes a write cycle to nonvolatile memory. Nonvolatile memory has a finite maximum number of write cycles. Programs that repeatedly cause write cycles to nonvolatile memory can eventually exceed the maximum number of write cycles and cause the memory to fail.

This command stores the present state of the instrument to the specified location in non-volatile memory. Up to 2 states can be stored - in locations 0 and 1. Any state previously stored in the same location will be overwritten. Use the *RCL command to retrieve instrument states. Refer to “*RST Settings” at the beginning of this chapter for a list of instrument settings that can be saved.

If a particular state is desired at power-on, it should be stored in location 0. It will then be automatically recalled at power turn-on if the Output Power-On state is set to RCL0.

Note that list data (and the calibration state) is not saved as part of the *SAV operation. This means that all list data that is sent to the instrument will be lost when the instrument is turned off.

Also, data saved in non-volatile memory, described in the “Non-volatile Factory Settings” table at the end of chapter 3 in the User’s Guide, is not affected by the *SAV command.

SRE**SRE?**

This command sets the value of the Service Request Enable Register. This register determines which bits from the Status Byte Register are summed to set the Master Status Summary (MSS) bit and the Request for Service (RQS) summary bit. A 1 in any Service Request Enable Register bit position enables the corresponding Status Byte Register bit. All such enabled bits are then logically ORed to cause the MSS bit (bit 6) of the Status Byte Register to be set.

When the controller conducts a serial poll in response to SRQ, the RQS bit is cleared, but the MSS bit is not. When *SRE is cleared (by programming it with 0), the instrument cannot generate an SRQ to the controller. The query returns the current state of *SRE.

***STB?**

This query reads the Status Byte register, which contains the status summary bits and the Output Queue MAV bit. Reading the Status Byte register does not clear it. The input summary bits are cleared when the appropriate event registers are read. The MAV bit is cleared at power-on, by *CLS, or when there is no more response data available.

A serial poll also returns the value of the Status Byte register, except that bit 6 returns Request for Service (RQS) instead of Master Status Summary (MSS). A serial poll clears RQS, but not MSS. When MSS is set, it indicates that the instrument has one or more reasons for requesting service.

Bit Position	7	6	5	4	3	2	1 – 0
Bit Value	128	64	32	16	8	4	–
Bit Name	OPER	MSS (RQS)	ESB	MAV	QUES	ERR	–

OPER = Operation status summary	MAV = Message available
MSS = Master status summary	QUES = Questionable status summary
(RQS) = Request for service	ERR = Error queue not empty
ESB = Event status byte summary	

***TRG**

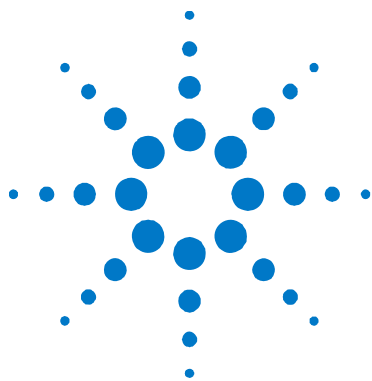
This command generates a trigger when the trigger subsystem has BUS selected as its source. The command has the same affect as the Group Execute Trigger (<GET>) command.

***TST?**

This query causes the instrument to do a self-test and report any errors. A 0 indicates the instrument passed self-test. A 1 indicates one or more tests failed. Selftest errors are written to the error queue (see appendix A). Note that *TST? also forces an *RST command.

***WAI**

This command instructs the instrument not to process any further commands until all pending operations are completed. Pending operations are as defined under the *OPC command. *WAI can be aborted only by sending the instrument a Device Clear command.



4

Programming Examples

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This chapter contains several example programs to help you develop programs for your own application. The example programs are for illustration only, and are provided with the assumption that you are familiar with the programming language being demonstrated and the tools used to create and debug procedures. See chapter 3, “Language Dictionary” for the SCPI command syntax.

You have a royalty-free right to use, modify, reproduce and distribute the example programs (and/or any modified version) in any way you find useful, provided you agree that Agilent Technologies has no warranty, obligations, or liability for any example programs.

The example programs are written in Microsoft Visual Basic 6.0 using the VISA COM IO library. The VISA COM library must be downloaded from the Automation-Ready CD-ROM to use these programs. For information about using VISA COM in another Visual Basic project, refer to “Programming Your Instruments” in the *USB/LAN/GPIB Interfaces Connectivity Guide*, also included on the Automation-Ready CD-ROM.

NOTE

Example programs for the following programming environments are also included on the Product-Reference CD-ROM located at the back of this guide:

- Microsoft Visual Basic 6.0
- Microsoft Visual C++ 6.0
- Microsoft Excel
- Agilent VEE Pro

The CD also contains IVI-COM and VXIplug&play drivers for your instrument.

Arb Programming Example

NOTE

This program only applies to the Agilent N6705A DC Power Analyzer

This program executes a voltage arbitrary waveform consisting of 10 steps. The steps are set to a dwell time of 1 second. When done, the program checks for instrument error and gives a message if there is an error. This is similar to the list programming example, but using SCPI Arb commands. Note that current values cannot be programmed as part of a voltage arbitrary waveform.

```
Sub arb_programming_example()
    Dim IDN As String
    Dim GPIBAddress As String
    Dim ErrString As String
    Dim channel As String

    'These variable are necessary to initialize the VISA COM.
    Dim ioMgr As AgilentRMLib.SRMCLs
    Dim Instrument As VisaComLib.FormattedIO488

    ' The following command line provides the program with the VISA name of the
    ' interface that it will communicate with. It is currently set to use GPIB.
    GPIBAddress = "GPIB1::5::INSTR"

    ' Use the following line instead for LAN communication
    ' TCPIPAddress="TCPIP0::141.25.36.214"

    ' Use the following line instead for USB communication
    ' USBAddress = "USB0::2391::1799::US00000002"

    ' Initialize the VISA COM communication
    Set ioMgr = New AgilentRMLib.SRMCLs
    Set Instrument = New VisaComLib.FormattedIO488
    Set Instrument.IO = ioMgr.Open(TCPIPAddress)

    ' These next three strings are the points in the user defined voltage arb.
    ' The first one controls voltage, the second dwell, and the third trigger out
    Const voltPoints = "1,2,3,4,5,6,7,8,9,10"
    Const dwellPoints = "1"
    Const triggerPoints = "0"

    ' This variable can be changed to program any channel in the mainframe
    channel = "(@1)" ' channel 1

    With Instrument
        ' Send a power reset to the instrument
        .WriteString "**RST"

        ' Query the instrument for the IDN string
        .WriteString "**IDN?"
        IDN = .ReadString

        ' Set the arb type to user defined voltage
        .WriteString "ARB:FUNC UDV," & channel

        ' Set the voltage mode to arb
        .WriteString "VOLT:MODE ARB," & channel
    End With
End Sub
```

```

' Send the voltage points
.WriteString "ARB:VOLT:UDEF:LEV " & voltPoints & "," & channel

' Send the dwell points
.WriteString "ARB:VOLT:UDEF:DWEL " & dwellPoints & "," & channel

'Send the trigger points
.WriteString "ARB:VOLT:UDEF:BOST " & triggerPoints & "," & channel

' Turn the output on
.WriteString "OUTP ON," & channel

' Wait for output to settle in the ON state.
.WriteString "*OPC?"
.ReadString

' Set the trigger source to bus
.WriteString "TRIG:TRAN:SOUR BUS," & channel

' Initiate the transient system
.WriteString "INIT:TRAN " & channel

' Trigger the unit
.WriteString "TRIG:TRAN " & channel

' Check instrument for any errors
.WriteString "Syst:err?"
ErrString = .ReadString

' give message if there is an error
If Val(ErrString) Then
    MsgBox "Error in instrument!" & vbCrLf & ErrString
End If
End With
End Sub

```

Output Programming Example

This is a simple program that sets a voltage, current, over-voltage, and the status of over-current protection. When done, the program checks for instrument error and gives a message if there is an error.

```
Sub output_programming_example()
    Dim IDN As String
    Dim GPIBAddress As String
    Dim ErrString As String

    ' This variable controls the channel number to be programmed
    Dim channel As String

    ' This variable controls the voltage
    Dim VoltSetting As Double

    ' This variable measures the voltage
    Dim MeasureVoltString As String

    ' This variable controls the current
    Dim CurrSetting As Double

    ' This variable controls the over voltage protection setting
    Dim overVoltSetting As Double

    'These variables are necessary to initialize the VISA COM.
    Dim ioMgr As AgilentRMLib.SRMCLs
    Dim Instrument As VisaComLib.FormattedIO488

    ' The following command line provides the program with the VISA name of the
    ' interface that it will communicate with. It is currently set to use GPIB.
    GPIBAddress = "GPIB0::5::INSTR"

    ' Use the following line instead for LAN communication
    ' TCPIPAddress="TCPIP0::141.25.36.214"

    ' Use the following line instead for USB communication
    ' USBAddress = "USB0::2391::1799::US000000002"

    ' Initialize the VISA COM communication
    Set ioMgr = New AgilentRMLib.SRMCLs
    Set Instrument = New VisaComLib.FormattedIO488
    Set Instrument.IO = ioMgr.Open(GPIBAddress)

    ' The next three command lines set the voltage, current, and over voltage
    VoltSetting = 3
    CurrSetting = 1.5                                ' amps
    overVoltSetting = 10

    ' This variable can be changed to program any channel in the mainframe
    channel = "(@1)"                                ' channel 1

    With Instrument
        ' Send a power reset to the instrument
        .WriteString "*RST"

        ' Query the instrument for the IDN string
        .WriteString "*IDN?"
        IDN = .ReadString
    End With
End Sub
```

```

' Set the voltage
.WriteString "VOLT" & Str$(VoltSetting) & "," & channel

' Set the over voltage level
.WriteString "VOLT:PROT:LEV " & Str$(overVoltSetting) & "," & channel

' Set current level
.WriteString "CURR " & Str$(CurrSetting) & "," & channel

' Turn on over current protection
.WriteString "CURR:PROT:STAT ON," & channel

' Turn the output on
.WriteString "OUTP ON," & channel

' Wait for previous commands to complete.
.WriteString "**OPC?"
.ReadString

' Measure the voltage
.WriteString "MEAS:VOLT? " & channel
MeasureVoltString = .ReadString
MsgBox "Measured Voltage is " & MeasureVoltString & "At channel" & channel

' Check instrument for any errors
.WriteString "Syst:err?"
ErrString = .ReadString

' give message if there is an error
If Val(ErrString) Then
    MsgBox "Error in instrument!" & vbCrLf & ErrString
End If
End With
End Sub

```

List Programming Example

This program executes a 10 point current and voltage list. It also specifies 10 different dwell times. When done, the program checks for instrument error and gives a message if there is an error.

```
Sub list_programming_example()
    Dim IDN As String
    Dim GPIBAddress As String
    Dim ErrString As String
    Dim channel As String

    'These variable are necessary to initialize the VISA COM.
    Dim ioMgr As AgilentRMLib.SRMCLs
    Dim Instrument As VisaComLib.FormattedIO488

    ' The following command line provides the program with the VISA name of the
    ' interface that it will communicate with. It is currently set to use GPIB.
    GPIBAddress = "GPIB1::5::INSTR"

    ' Use the following line instead for LAN communication
    ' TCPIPAddress="TCPIP0::141.25.36.214"

    ' Use the following line instead for USB communication
    ' USBAddress = "USB0::2391::1799::US00000002"

    ' Initialize the VISA COM communication
    Set ioMgr = New AgilentRMLib.SRMCLs
    Set Instrument = New VisaComLib.FormattedIO488
    Set Instrument.IO = ioMgr.Open(GPIBAddress)

    ' These next three strings are the points in the list.
    ' All three strings are the same length.
    ' The first one controls voltage, the second current, and the third dwell time
    Const voltPoints = "1,2,3,4,5,6,7,8,9,10"
    Const currPoints = "0.5,1,1.5,2,2.5,3,3.5,4,4.5,5"
    Const dwellPoints = "1,2,0.5,1,0.25,1.5,0.1,1,0.75,1.2"

    ' This variable can be changed to program any channel in the mainframe
    channel = "(@1)" ' channel 1

    With Instrument
        ' Send a power reset to the instrument
        .WriteString "*RST"

        ' Query the instrument for the IDN string
        .WriteString "*IDN?"
        IDN = .ReadString

        ' Set the voltage mode to list
        .WriteString "VOLT:MODE LIST," & channel

        ' Set the current mode to list
        .WriteString "CURR:MODE LIST," & channel

        ' Send the voltage list points
        .WriteString "LIST:VOLT " & voltPoints & "," & channel

        ' Send the Current list points
        .WriteString "LIST:CURR " & currPoints & "," & channel
    End With
End Sub
```

```

' Send the dwell points
.WriteString "LIST:DWEL " & dwellPoints & "," & channel

' Turn the output on
.WriteString "OUTP ON," & channel

' Wait for output to settle in the ON state.
.WriteString "*OPC?"
.ReadString

' Set the trigger source to bus
.WriteString "TRIG:TRAN:SOUR BUS," & channel

' Initiate the transient system
.WriteString "INIT:TRAN " & channel

' Trigger the unit
.WriteString "*TRG"

' Check instrument for any errors
.WriteString "Syst:err?"
ErrString = .ReadString

' give message if there is an error
If Val(ErrString) Then
    MsgBox "Error in instrument!" & vbCrLf & ErrString
End If
End With
End Sub

```

Digitizer Programming Example

This program uses the voltage in step mode and also demonstrates how to set up and use the digitizer. When done, the program checks for instrument error and gives a message if there is an error.

```
Sub digitizer_programming_example()
    Dim IDN As String
    Dim GPIBAddress As String
    Dim ErrString As String
    Dim channel As String
    Dim measPoints As Long
    Dim measOffset As Long
    Dim VoltSetting As Double
    Dim finalVoltage As Double
    Dim timeInterval As Double
    Dim VoltPoints() As Variant
    Dim i As Long
    Const WTG_MEAS = 8
    Const WTG_TRANS = 16

    'These variables are necessary to initialize the VISA COM.
    Dim ioMgr As AgilentRMLib.SRMCLs
    Dim Instrument As VisaComLib.FormattedIO488

    ' The following command line provides the program with the VISA name of the
    ' interface that it will communicate with. It is currently set to use GPIB.
    GPIBAddress = "GPIB0::5::INSTR"

    ' Use the following line instead for LAN communication
    ' TCPIPAddress="TCPIP0::141.25.36.214"

    ' Use the following line instead for USB communication
    ' USBAddress = "USB0::2391::1799::US000000002"

    ' Initialize the VISA COM communication
    Set ioMgr = New AgilentRMLib.SRMCLs
    Set Instrument = New VisaComLib.FormattedIO488
    Set Instrument.IO = ioMgr.Open(GPIBAddress)

    ' This controls the number of points the measurement system measures
    measPoints = 100

    ' This controls the number of points to offset the measurement (positive for
    ' forward, negative for reverse)
    measOffset = 0

    ' This sets the time between points
    timeInterval = 0.0025

    ' This controls the voltage
    VoltSetting = 5

    ' This is the final voltage that will be triggered
    finalVoltage = 10

    ' This variable can be changed to program any channel in the mainframe
    channel = "@1"                ' channel 1
```

```

With Instrument
' Send a power reset to the instrument
.WriteString "*RST"

' Query the instrument for the IDN string
.WriteString "*IDN?"
IDN = .ReadString

' Put the Voltage into step mode which causes it to transition from one
' voltage to another upon receiving a trigger
.WriteString "VOLT:MODE STEP," & channel

' Set the voltage
.WriteString "VOLT" & Str$(VoltSetting) & "," & channel

' Go to final value
.WriteString "VOLT:TRIG" & Str$(finalVoltage) & "," & channel

' Turn the output on
.WriteString "OUTP ON," & channel

' Wait for output to turn on.
.WriteString "*OPC?"
.ReadString

' Set the bus as the transient trigger source
.WriteString "TRIG:TRAN:SOUR BUS," & channel

' Set the number of points for the measurement system to use as an offset
.WriteString "SENS:SWE:OFFS:POIN" & Str$(measOffset) & "," & channel

' Set the number of points that the measurement system uses
.WriteString "SENS:SWE:POIN" & Str$(measPoints) & "," & channel

' Set the time interval between points
.WriteString "SENS:SWE:TINT" & Str$(timeInterval) & "," & channel

' Set the measurement trigger source
.WriteString "TRIG:ACQ:SOUR BUS," & channel

' Initiate the measurement trigger system
.WriteString "INIT:ACQ " & channel

' Initiate the transient trigger system
.WriteString "INIT:TRAN " & channel

' Wait for INITiate commands to complete
Do
    .WriteString "STAT:OPER:COND? " & channel
    Status = .ReadString
Loop Until (Status And (WTG_MEAS + WTG_TRANS)) = (WTG_MEAS + WTG_TRANS)

' Trigger the unit
.WriteString "*TRG"

' Read back the voltage points
.WriteString "FETC:ARR:VOLT? " & channel
VoltPoints = .ReadList

' Print the first 10 voltage points
For i = 0 To 9
    Debug.Print i, VoltPoints(i)
Next i

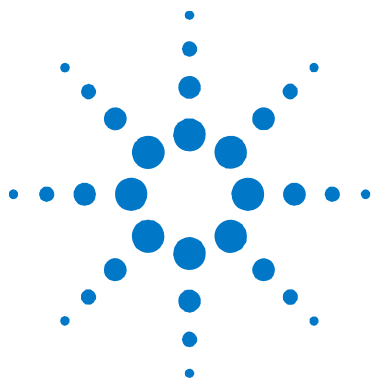
```

4 Programming Examples

```
' Check instrument for any errors
.WriteString "Syst:err?"
ErrString = .ReadString

' give message if there is an error
If Val(ErrString) Then
    MsgBox "Error in instrument!" & vbCrLf & ErrString
End If
End With

End Sub
```



Appendix A

Error Messages

Displaying Error Messages	90
Error List	90

This appendix gives the error numbers and descriptions that are returned by the Agilent N6700 and N6705A.

Displaying Error Messages

The instrument beeps each time a command syntax or hardware error is generated. The front-panel **ERR** annunciator turns on when one or more errors are currently stored in the SCPI error queue.

Front Panel:	SCPI Command:
Press the Error key.	SYST : ERR?
If errors appear, use the navigation keys to scroll through the list.	

Errors are cleared as they are read. When all errors have been read, the **ERR** annunciator turns off and the error queue is cleared.

Error List

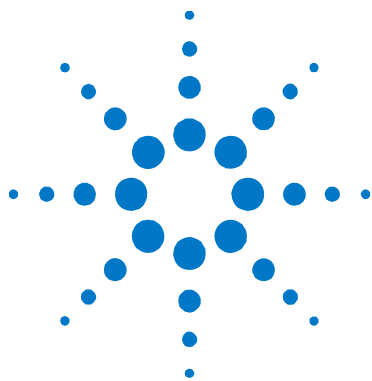
Error	Device-dependent Errors (these errors set Standard Event Status register bit #3)
0	No error This is the response to the ERR? query when there are no errors.
100	Too many channels You have specified more channels than are installed in the mainframe.
101	Calibration state is off Calibration is not enabled. The instrument will not accept calibration commands.
102	Calibration password is incorrect The calibration password is incorrect.
103	Calibration is inhibited by switch setting Calibration mode is locked out by the calibration switch.
104	Bad sequence of calibration commands Calibration commands have not been entered in the proper sequence.
105	Unexpected output current The measured output current is outside the acceptable range.
106	Zero measurement out of range error The "zero" measurement value is outside the acceptable range.
107	Programming cal constants out of range The programmed calibration constant is outside the acceptable range.
108	Measurement cal constants out of range The measurement calibration constant is outside the acceptable range.
109	Over voltage cal constants out of range The over voltage calibration constant is outside the acceptable range.
110	Wrong V+I The instrument was unable to set the correct voltage or current value.
111	Aux vloc cal constants out of range Calibration constants on the internal auxiliary local ADC are outside the acceptable range.
112	Aux vrem cal constants out of range Calibration constants on the internal auxiliary remote ADC are outside the acceptable range.
113	Aux imon cal constants out of range Calibration constants on the internal auxiliary imon ADC are outside the acceptable range.

Device-dependent Errors (continued)	
200	Hardware error channel <channel> A hardware error has occurred on the specified channel.
201	Invalid configuration, empty slots There is an empty slot between modules. This configuration is not allowed.
202	Selftest Fail A selftest failure has occurred. See selftest failure list for details.
203	Compatibility function not implemented The requested compatibility function is not available.
204	NVRAM checksum error A checksum error has occurred in the instrument's nonvolatile random access memory.
205	NVRAM full The nonvolatile random access memory of the instrument is full.
206	File not found The internal calibration file or the internal channel attribute file was not found in NVRAM.
207	Cal file version error The calibration file was written or read using old firmware. Firmware must be updated.
302	Option not installed The option that is programmed by this command is not installed.
303	There is not a valid acquisition to fetch from There is no valid data in the measurement buffer.
304	Volt and curr in incompatible transient modes Voltage and current cannot be in Step and List mode at the same time.
305	A triggered value is on a different range A triggered value is on a different range than the one that is presently set.
306	Too many list points Too many list points have been specified.
307	List lengths are not equivalent One or more lists are not the same length.
308	This setting cannot be changed while transient trigger is initiated Setting cannot be changed while the instrument is waiting for or executing a trigger sequence.
309	Cannot initiate, voltage and current in fixed mode Cannot initiate the transient generator because either the voltage or the current function is set to Fixed mode.
Command Errors (these errors set Standard Event Status register bit #5)	
-100	Command error Generic syntax error.
-101	Invalid character An invalid character was found in the command string.
-102	Syntax error Invalid syntax was found in the command string. Check for blank spaces.
-103	Invalid separator An invalid separator was found in the command string. Check for proper use of , ; :
-104	Data type error A different data type than the one allowed was found in the command string.
-105	GET not allowed A group execute trigger is not allowed in a command string.

Command Errors (continued)	
–108	Parameter not allowed More parameters were received than were expected.
–109	Missing parameter Fewer parameters were received than were expected.
–110	Command header error An error was detected in the header.
–111	Header separator error A character that was not a valid header separator was found in the command string.
–112	Program mnemonic too long The header contains more than 12 characters.
–113	Undefined header A command was received that was not valid for this instrument.
–114	Header suffix out of range The value of the numeric suffix is not valid.
–120	Numeric data error Generic numeric data error.
–121	Invalid character in number An invalid character for the data type was found in the command string.
–123	Exponent too large The magnitude of the exponent was larger than 32000.
–124	Too many digits The mantissa of a numeric parameter contained more than 255 digits, excluding leading zeros.
–128	Numeric data not allowed A numeric parameter was received but a character string was expected.
–130	Suffix error Generic suffix error
–131	Invalid suffix A suffix was incorrectly specified for a numeric parameter.
–134	Suffix too long The suffix contains more than 12 characters.
–138	Suffix not allowed A suffix is not supported for this command.
–140	Character data error Generic character data error
–141	Invalid character data Either the character data element contains an invalid character, or the element is not valid.
–144	Character data too long The character data element contains more than 12 characters.
–148	Character data not allowed A discrete parameter was received, but a string or numeric parameter was expected.
–150	String data error Generic string data error
–151	Invalid string data An invalid character string was received. Check that the string is enclosed in quotation marks.
–158	String data not allowed A character string was received, but is not allowed for this command.

Command Errors (continued)	
–160	Block data error Generic block data error
–161	Invalid block data The number of data bytes sent does not match the number of bytes specified in the header.
–168	Block data not allowed Data was sent in arbitrary block format but is not allowed for this command.
–170	Expression error Generic expression error
–171	Invalid expression data The expression data element was invalid.
–178	Expression data not allowed Expression data element was sent but is not allowed for this command.
Execution Errors (these errors set Standard Event Status register bit #4)	
–200	Execution error Generic syntax error
–220	Parameter error A data element related error occurred.
–221	Settings conflict A data element could not be executed because of the present instrument state.
–222	Data out of range A data element could not be executed because the value was outside the valid range.
–223	Too much data A data element was received that contains more data than the instrument can handle.
–224	Illegal parameter value An exact value was expected but not received.
–225	Out of memory The device has insufficient memory to perform the requested operation.
–226	Lists not same length One or more lists are not the same length.
–230	Data corrupt or stale Possible invalid data. A new reading was started but not completed.
–231	Data questionable The measurement accuracy is suspect.
–232	Invalid format The data format or structure is inappropriate.
–233	Invalid version The version of the data format is incorrect to the instrument.
–240	Hardware error The command could not be executed because of a hardware problem with the instrument.
–241	Hardware missing The command could not be executed because of missing hardware, such as an option.
–260	Expression error An expression program data element related error occurred.
–261	Math error in expression An expression program data element could not be executed due to a math error.

Query Errors (these errors set Standard Event Status register bit #2)	
-400	Query Error Generic error query
-410	Query INTERRUPTED A condition causing an interrupted query error occurred.
-420	Query UNTERMINATED A condition causing an unterminated query error occurred.
-430	Query DEADLOCKED A condition causing a deadlocked query error occurred.
-440	Query UNTERMINATED after indefinite response A query was received in the same program message after a query indicating an indefinite response was executed.
Selftest Errors (these errors set Standard Event Status register bit #3)	
202	Selftest Fail Aux Adc 0 expected <n1> to <n2>, measured <n3>, chan <n4> Auxiliary ADC failed. n1 and n2 are the expected limits. n3 is the measured value. n4 is the channel location of the failed module.
202	Selftest Fail DACs 0 expected <n1> to <n2>, measured <n3>, chan <n4> Both voltage and current DACs are at zero. n1 and n2 are the expected limits. n3 is the measured value. n4 is the channel location of the failed module.
202	Selftest Fail DACs 1 expected <n1> to <n2>, measured <n3>, chan <n4> Voltage DAC is at zero; current DAC is at full scale. n1 and n2 are the expected limits. n3 is the measured value. n4 is the channel location of the failed module.
202	Selftest Fail DACs 2 expected <n1> to <n2>, measured <n3>, chan <n4> Voltage DAC is at full scale; current DAC is at zero. n1 and n2 are the expected limits. n3 is the measured value. n4 is the channel location of the failed module.
202	Selftest Fail DACs 3 expected <n1> to <n2>, measured <n3>, chan <n4> Both voltage and current DACs are at full scale. n1 and n2 are the expected limits. n3 is the measured value. n4 is the channel location of the failed module.



Appendix B Compatibility

Differences – In General	96
Compatibility Command Summary	97

The Agilent N6700 Modular Power System is programmatically compatible with the Agilent 6621A – 6629A DC power supplies. This means that you can program the power system using the same commands that are used to program the Agilent 662xA power supplies.

Agilent 662xA commands will only work in mainframes that contain combinations of Agilent N675xA or N676xA power modules.

NOTE

Agilent 662xA commands will NOT work in mainframes that contain any Agilent of N673xA/B, N674xA/B, or N677xA power modules. This is true even if the mainframes also contain Agilent N675xA or N676xA power modules.

Differences – In General

The following table documents the general differences between the way Compatibility programming commands work on the Agilent N6700 Modular Power System and the way they work on the Agilent 662xA series DC power supplies.

Item	Differences
Queries	The Agilent N67xxA modules will respond to multiple queries. It will not allow a space separator between numbers. It will not allow a user to query information, read back only a portion of the information, send another command, and finish reading back the information from the original query. Sending a second query without reading the response to the first will generate an error. Model number queries will only return the N6700 model number.
Status functions	Serial Poll will be controlled by the SCPI status model and will not act like a 662xA serial poll. SRQ will be controlled by the SCPI status model. Parallel poll will not work. The Agilent N67xxA modules will never set the CC- bit. All other status bits are translated to the closest 662xA bits.
Settings	The full-scale limits will match the Agilent N67xxA module limits. There is no automatic setting adjustment.
Measurement	Measurements use the present settings for points, time interval, etc. Floating point numbers returned by the power system may not have exactly the same syntax or number of digits.
Calibration	Calibration must be done in SCPI.
Storage states	The Agilent N67xxA modules have 2 non-volatile states and 9 volatile states. The 9 volatile states are only available when using compatibility commands.
Protection functions	OVRST and OCRST will reset all latched protection functions, not just over-voltage or over-current.

Compatibility Command Summary

The following table documents the compatibility commands that the Agilent N6700 MPS supports. All compatibility commands are accepted; however, some commands will generate an error.

Agilent 662xA Command	Description	Similar SCPI Command
ASTS? <ch>	Queries the accumulated status (ASTS) of the specified output channel. The response represents the sum of the binary weights of the ASTS register bits. The ASTS register is automatically set to the present status after being queried.	STAT:OPER:EVEN? STAT:QUES:EVEN? *ESE?
CLR	Returns the entire power system (all outputs) to the power-on state, except the unit is not unaddressed and its store/recall registers are not changed.	*RST
CMODE <on/off>	Generates error –113. The unit must be calibrated using SCPI commands.	see Appendix B
CMODE?	Generates error –113. The unit must be calibrated using SCPI commands.	see Appendix B
DC <ch>, < on/off >	Turns the specified output channel on or off. On/off = 0 turns the channel off; on/off = 1 turns the channel on.	OUTP:STAT
DC? <ch>	Queries whether the specified output channel is on or off. Response is either 1 (on) or 0 (off).	OUTP:STAT?
DCPON < on/off >	Sets the power-on state of the output channels. For on/off = 0, all outputs will be off state when the unit is turned on. For on/off = 1 all outputs will be in the on state when the power system is turned on. This command requires the output power-on state to be set to RCL0.	OUTP:PON:STAT RCL0
DCPON?	Returns a 1 if any channels are set to turn on their output at power on.	OUTP:PON:STAT?
DLY <ch>, <delay>	Sets the reprogramming delay for the specified output channel. This delay is used to mask the CV, CC, CP-, UNR, and other status bits from the fault register and the OCP function for the specified delay period.	OUTP:PROT:DEL
DLY? <ch>	Queries the present reprogramming delay of the specified output channel.	OUTP:PROT:DEL?
DSP <on/off>	Does nothing.	None
DSP?	Always returns a 1.	None
DSP "xxxxxxxxxxx"	Does nothing.	None
ERR?	Queries the present programming or hardware error. An error code number is returned over the GPIB to identify the error.	SYST:ERR?
FAULT? <ch>	Queries the fault register of the specified output channel. A bit is set in the fault register when the corresponding bit in both the status and the mask registers. The response is an integer 0 to 255. The fault register is cleared after being read.	STAT:OPER? STAT:QUES? *ESE?
ID?	Queries the model number of the power system.	*IDN?
IDATA <ch>, <llo>, <lhi>	Generates error –113. Unit must be calibrated using SCPI commands.	see Appendix B

Agilent 662xA Command	Description	Similar SCPI Command
IHI <ch>	Generates error –113. Unit must be calibrated using SCPI commands.	see Appendix B
ILO <ch>	Generates error –113. Unit must be calibrated using SCPI commands.	see Appendix B
IOUT? <ch>	Queries the measured output current of the specified output channel. The response is a real number. The front panel display can be used to monitor the measured output current (and voltage) of the selected output channel.	MEAS:CURR?
IRLO <ch>	Generates error –113.	None
IRHI <ch>	Generates error –113.	None
IRLN <ch>	Generates error –113.	None
IRHN <ch>	Generates error –113.	None
IRSET <ch>, <x>	Sets the full-scale current range programming for an output channel. The power system will select the lowest operating range that will contain the value (x). If the present operating current is greater than the full-scale range selected, the present output current will be reduced to the full-scale value of the new range selected. If the current entered is larger than the highest possible range, an error will result.	SENS:CURR:RANG
IRSET? <ch>	Queries the full-scale current range setting of an output channel.	SENS:CURR:RANG?
ISSET <ch>, <current>	Sets the current of the specified output channel.	CURR
ISSET? <ch>	Queries the present current setting of the specified output channel. The response is a real number.	CURR?
ISTEP <ch>, <x>	Increases or decreases the output channel current in steps. If the step value < x > is positive, output current will increase by the step value, if the step value is negative, the output current will decrease by the step value. Stepping beyond the active operating range causes an error.	LIST:CURR
METER <ch>	Does nothing.	None
METER?	Always returns a 1.	None
OCP <ch>, <on/off>	Enables the over-current protection circuit for the specified output channel. This circuit, when enabled, causes the output to go to the off state when the output is in + CC mode. On/off is a 1 to turn on (enable) or a 0 to turn off (disable) the circuit.	CURR:PROT:STAT
OCP? < ch >	Queries the over-current protection circuit on/off status for the specified output channel. Response is either a 1 (on) or a 0 (off).	CURR:PROT:STAT?
OCRST < ch >	Returns the specified output channel to its previous settings after it has been turned off by a protection shutdown.	OUTP:PROT:CLE
OUT < ch >, < on/off >	Turns the specified output channel on or off. <on/off> equals 1 to turn the output on and equals 0 to turn the output off.	OUTP
OUT? < ch >	Queries whether the specified output channel is turned on or off. The response is either 1 (on) or 0 (off).	OUTP?

Agilent 662xA Command	Description	Similar SCPI Command
OVCAL < ch >	Generates error –113. Unit must be calibrated using SCPI commands.	see Appendix B
OVRST < ch >	Returns the specified output channel to its previous settings after it has been turned off by a protection shutdown.	OUTP:PROT:CLE
OVSET <ch>, <ovpoint>	Sets the over-voltage trip point for the specified output channel.	VOLT:PROT:LEV
OVSET? < ch >	Queries the present over-voltage setting of the specified output channel. The response is a real number.	VOLT:PROT:LEV?
PON < on/off >	Does nothing.	*SRE
PON?	Always returns a 0.	*SRE?
RCL < reg >	Recalls the settings for all output channels from the specified internal register. These settings were previously stored using the STO command. RCL programs all output channels in sequential order to these stored settings.	*RCL
RELAY < on/off >	Generates error 203. This Option 750 capability is not supported.	None
RELAY?	Generates error 203. This Option 750 capability is not supported.	None
RIRST < ch >	Resets all protection functions for the specified channel. This Option 750 capability is supported	OUTP:PROT:CLE
ROM?	Queries the revision of the power system's firmware.	*IDN?
SROM?	Queries the revision of the power system's firmware.	*IDN?
SRQ < setting >	Does nothing.	*SRQ
SRQ?	Always returns a 0.	*SRQ?
STO <reg>	Stores the present settings for all output channels in the specified register. These settings can be recalled when desired.	*SAV
STS? <ch>	Queries the present status of the specified output channel. The response represents the sum of the binary weights of the status register bits.	STAT:OPER:COND? STAT:QUES:COND?
TEST?	Always returns a 0.	*TST?
UNMASK <ch>, <setting>	Sets the bits in the mask register of the specified output channel to the setting. The setting is an integer that represents the sum of the binary weights of the bits. The mask register operates in conjunction with the status and fault registers.	STAT:OPER:NTR STAT:OPER:PTR STAT:QUES:NTR STAT:QUES:PTR
UNMASK? <ch>	Queries the present setting of the mask register of the specified output channel. The response is an integer from 0 to 255.	STAT:OPER:NTR? STAT:OPER:PTR? STAT:QUES:NTR? STAT:QUES:PTR?
VDATA <ch>, <Vlo>, <Vhi>	Generates error –113. Unit must be calibrated using SCPI commands.	see Appendix B
VHI <ch>	Generates error –113.	None
VLO <ch>	Generates error –113.	None

Agilent 662xA Command	Description	Similar SCPI Command
VMUX? <ch>, <input>	Generates error –113.	None
VOUT? <ch>	Queries the measured output voltage of the specified output channel. The response is a real number. The front panel display can be used to monitor the measured output voltage (and current) of the selected output channel.	MEAS:VOLT?
VRLO <ch>	Generates error –113.	None
VRHI <ch>	Generates error –113.	None
VRSET <ch>, <x>	Sets the full-scale voltage range programming for an output channel. The power system will select the lowest operating range that will contain the value (x). If the present operating voltage is greater than the full-scale range selected, the present output voltage will be reduced to the full-scale value of the new range selected. If the voltage entered is larger than the highest possible range, an error message will result.	None
VRSET? <ch>	Queries the full scale voltage range setting of an output channel.	None
VSET <ch>, <voltage>	Sets the voltage of the specified output channel.	VOLT
VSET? <ch>	Queries the present voltage setting of the specified output channel. The response is a real number.	VOLT?
VSTEP <ch>, <x>	Increases or decreases the output channel voltage in steps. If the step value < x > is positive, output voltage will increase by the step value, if the step value is negative, the output voltage will decrease by the step value. Trying to step beyond the operating range will cause a number range error.	LIST:VOLT

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